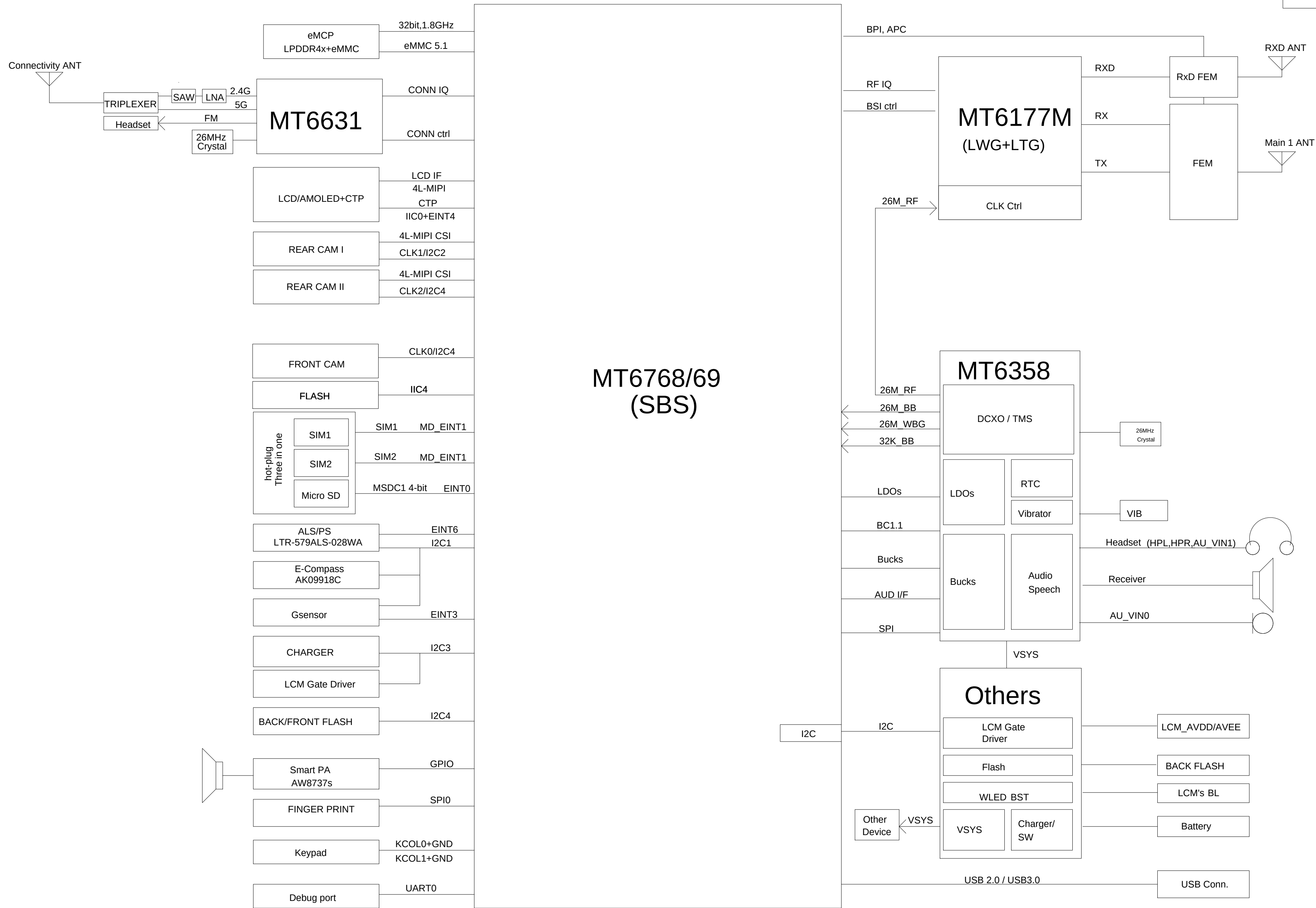


BLOCK_DIAGRAM

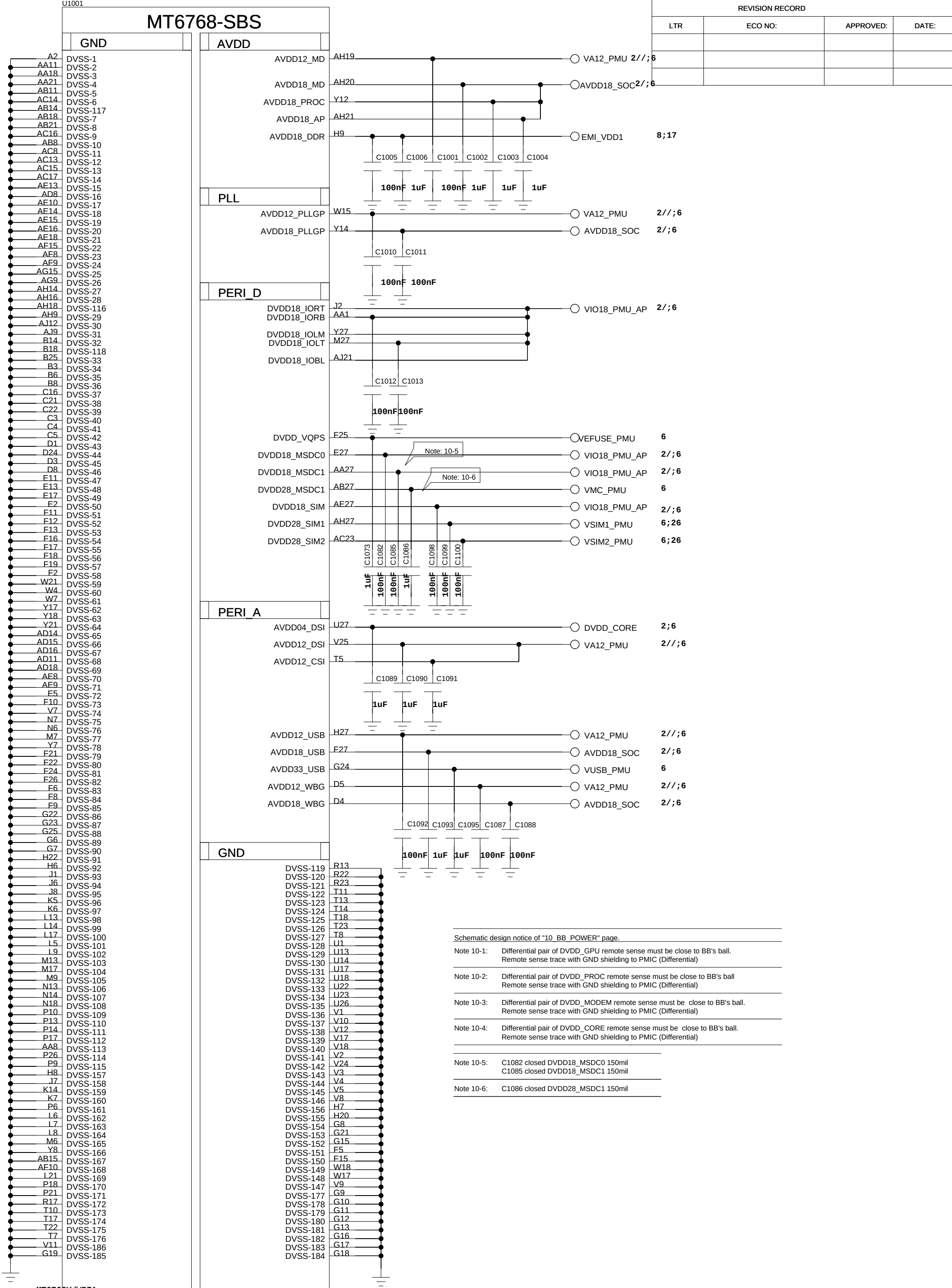
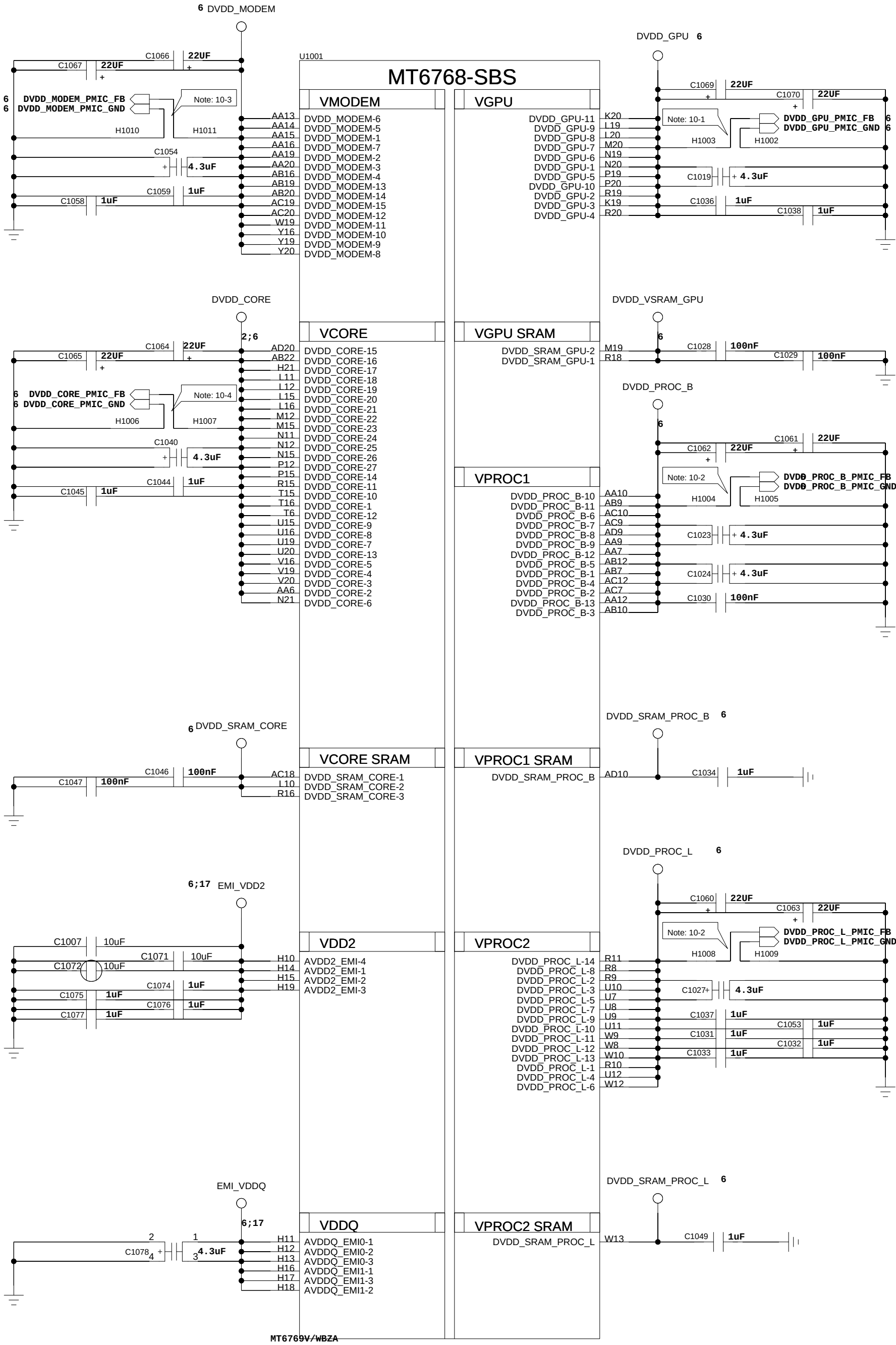
Project : MT6768/69 LPDDR4x

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



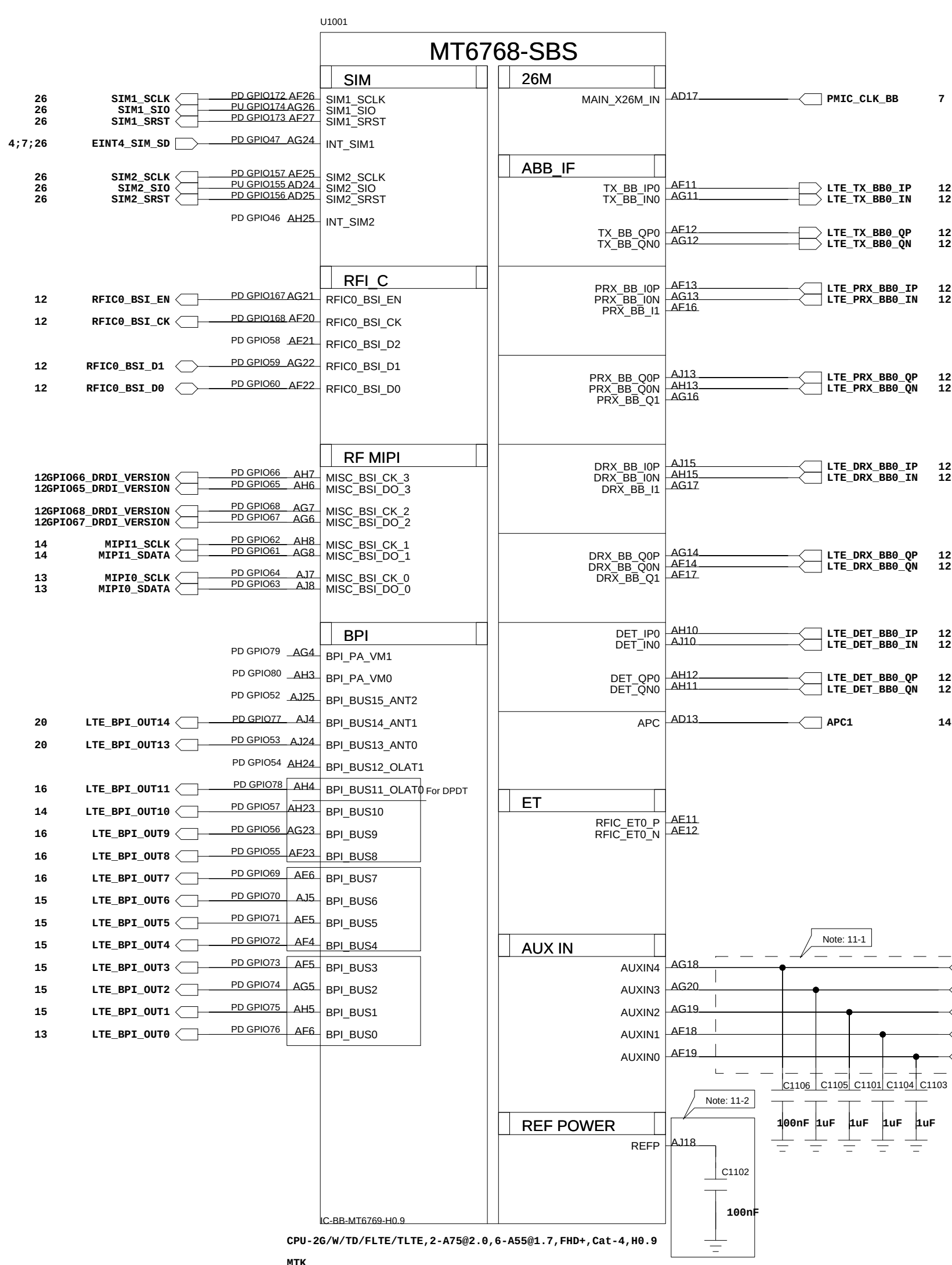
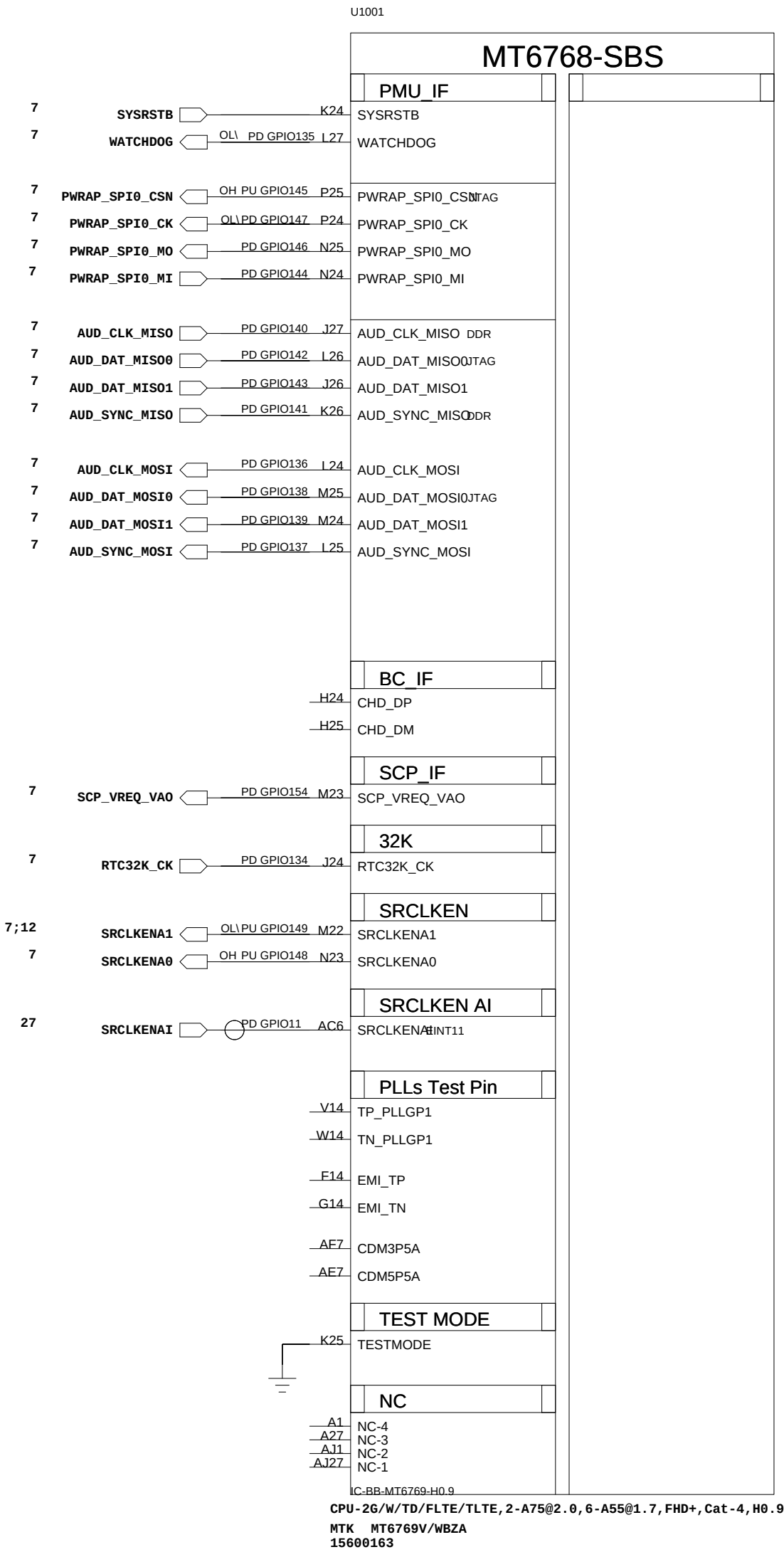
COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 00_BLOCK_DIAGRAM		VERSION:	SHEET: 1 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

BB_POWER



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 10_BB_POWER		VERSION:	SHEET: 2 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



"PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.			
	PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG
Default	HI	LO	N/A
	HI	HI	SPI0+EINT8
	LO (No test)	LO (No test)	SPI0+EINT8
	LO (No test)	HI (No test)	N/A

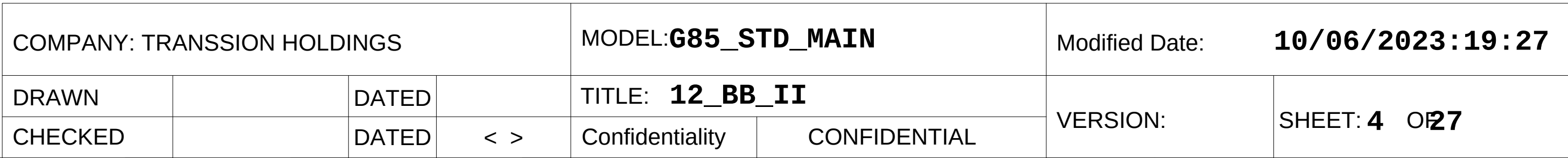
AUD_SYNC_MISO and AUD_CLK_MISO are DDR type feature in bootstrap			
	AUD_SYNC_MISO	AUD_CLK_MISO	DDR Type
Default	LO	LO	LP4X eMCP
	LO	HI (No test)	Reserved
	HI (No test)	LO	LP3 eMCP
	HI (No test)	HI (No test)	Reserved

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 11_BB_I		VERSION:	SHEET: 3 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

Schematic design notice of "11_BB_I" page.

Note 11-1: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

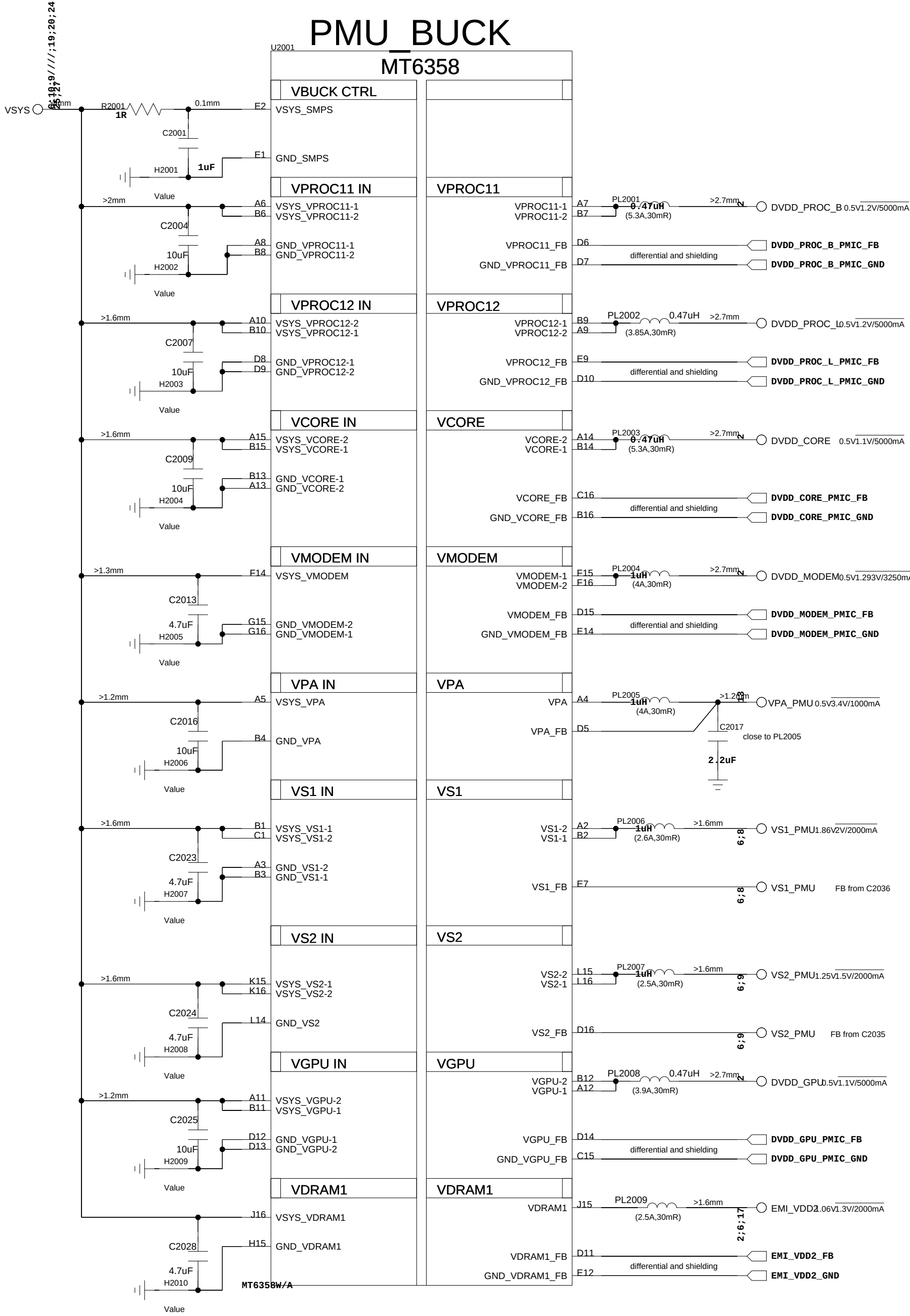
Note 11-2: The de-coupling cap. for REFP (AJ21 ball) have to be placed as close to BB as possible.



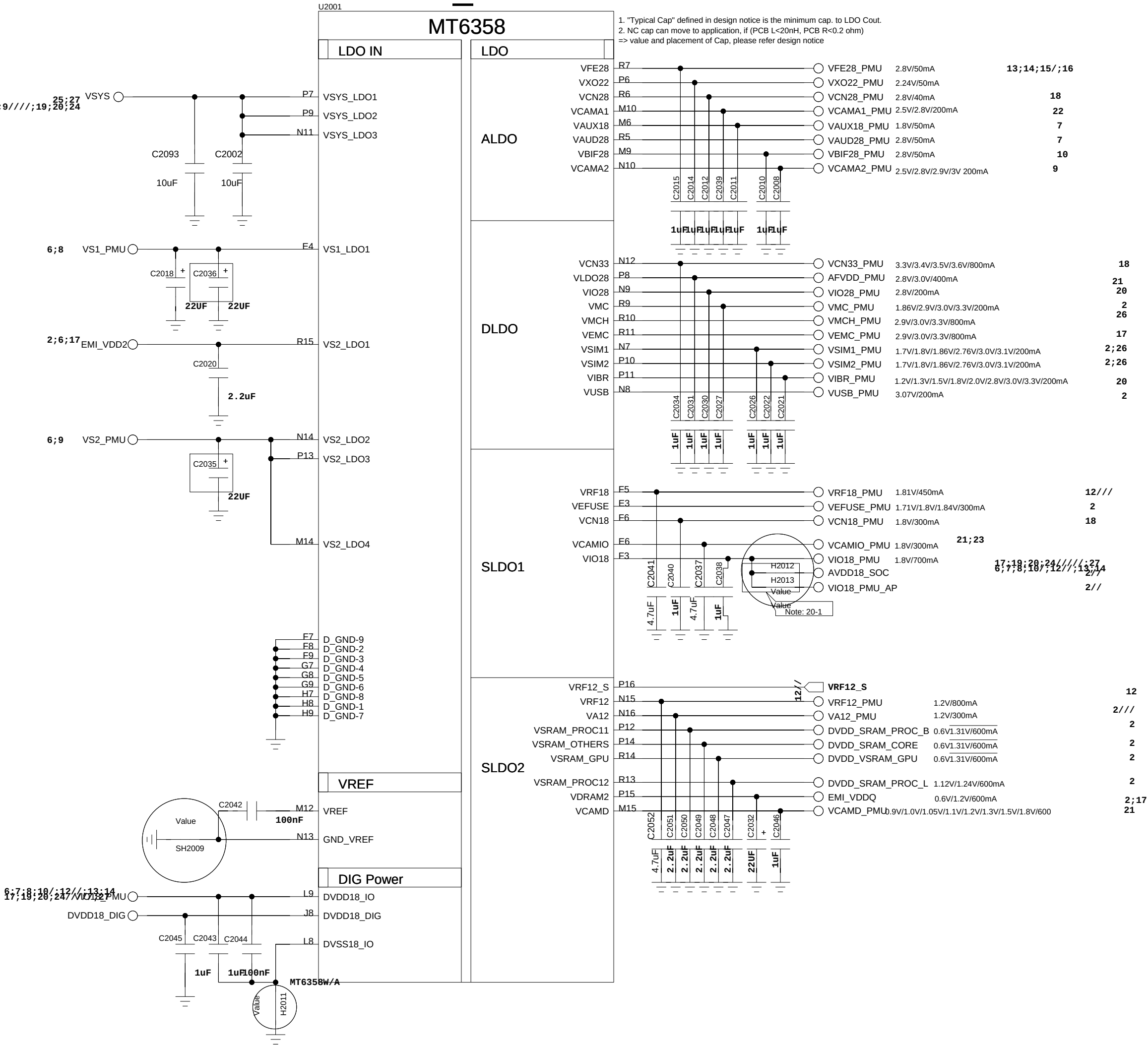
POWER_MT6358_I

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PMU BUCK



PMU LDO



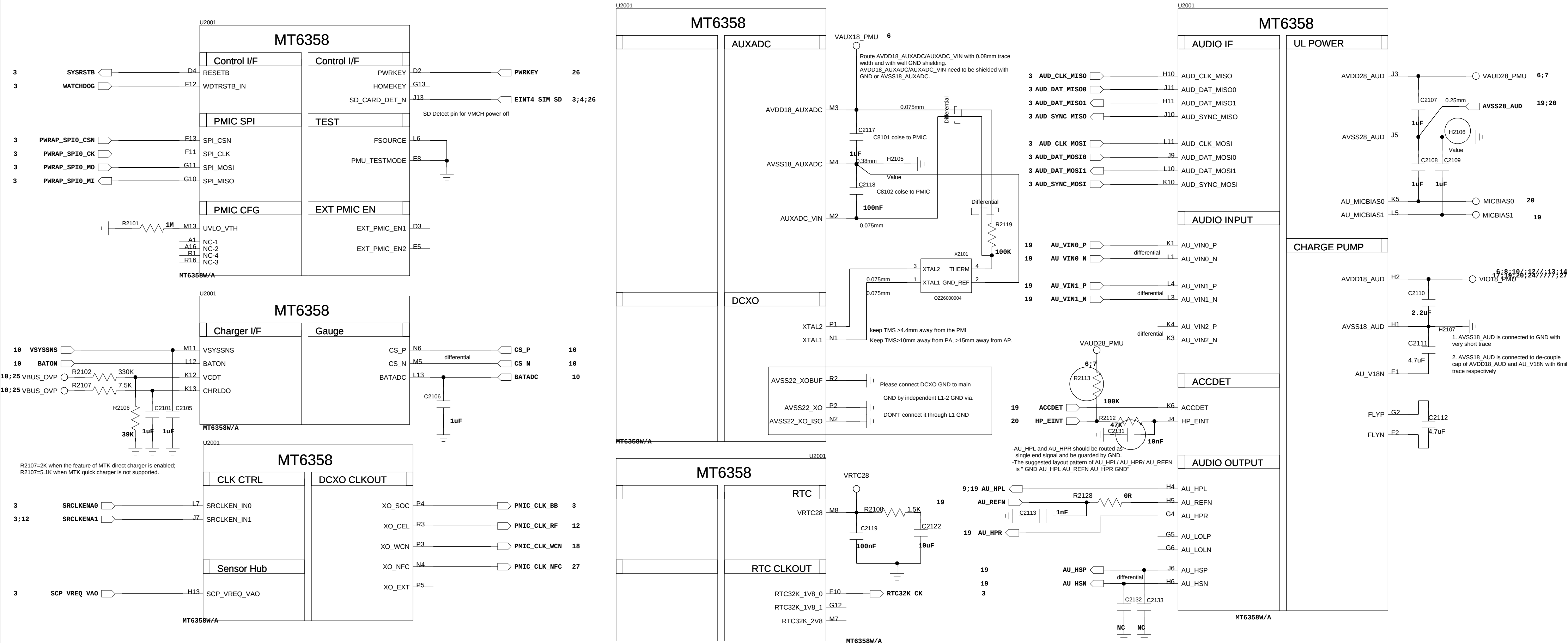
Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set H2012/H2013 close to C2038, making star connection between VIO18_PMU and AVDD18_SOC/VIO18_PMU_AP near to LDO cap. C2038
Please also refer to MT6358 design notice for further detail design information

COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 20_POWER_MT6358_I		VERSION:	SHEET: 6 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_MT6358_II

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

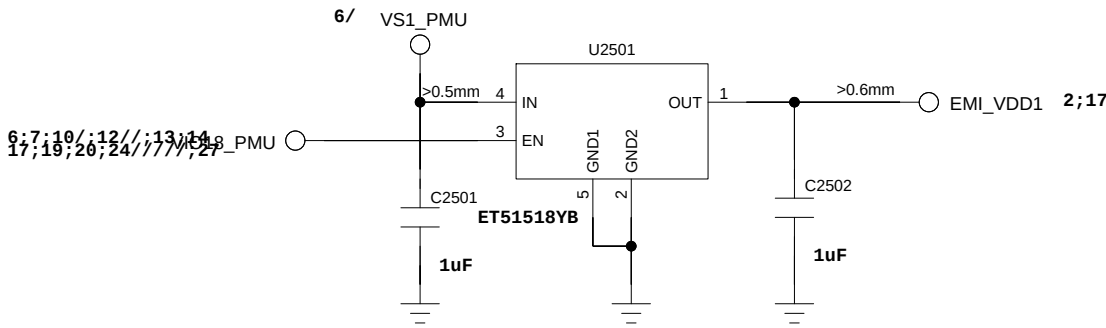


COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 21_POWER_MT6358_II		VERSION:	SHEET: 7 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_THIRD-PARTY-I

System supply

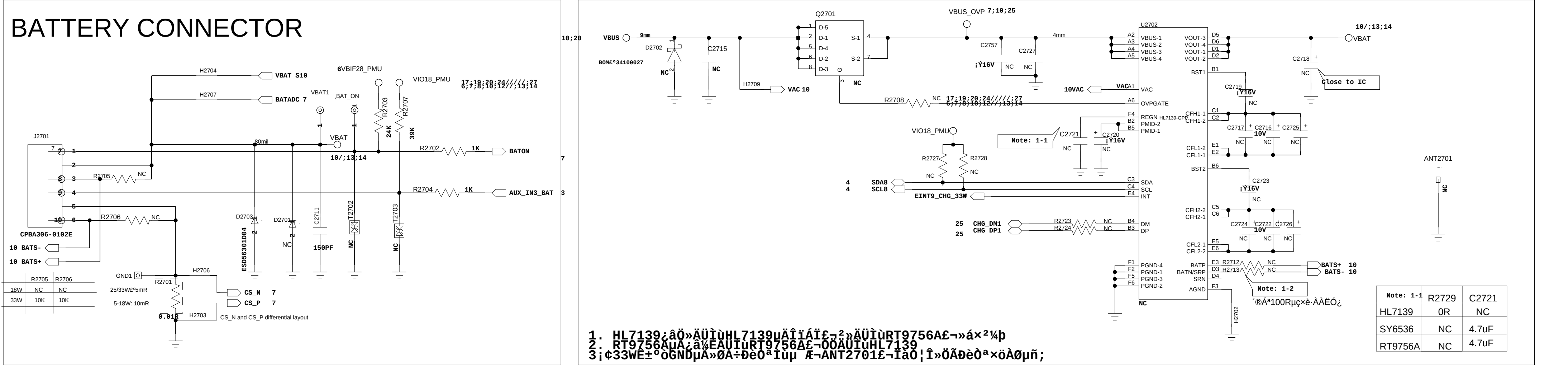
LPDDR4X VDD1 1.8V LDO



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 25_POWER_THIDR-PARTY_I		VERSION:	SHEET: 8 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 27_POWER_THIRD-PARTY_III		VERSION:	SHEET: 10 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_THIRD-PARTY-IV

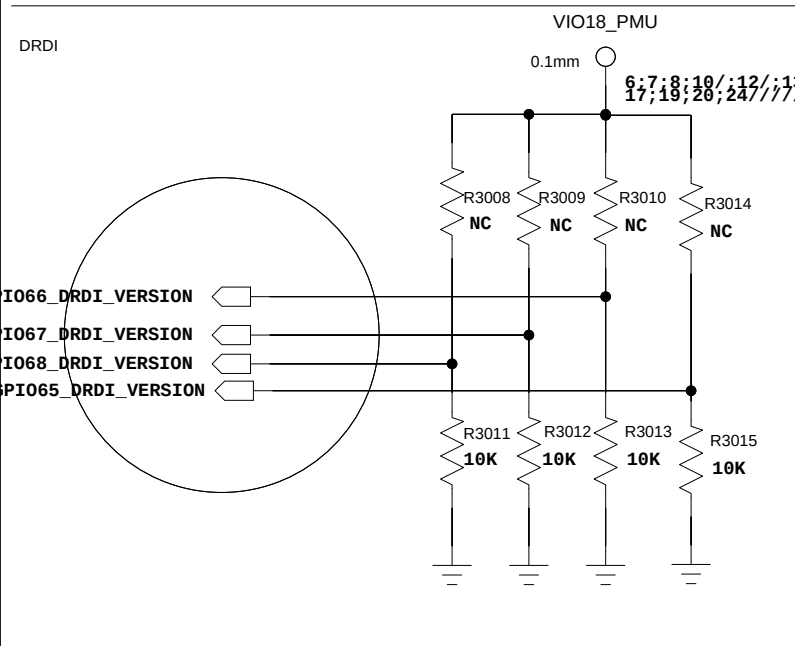
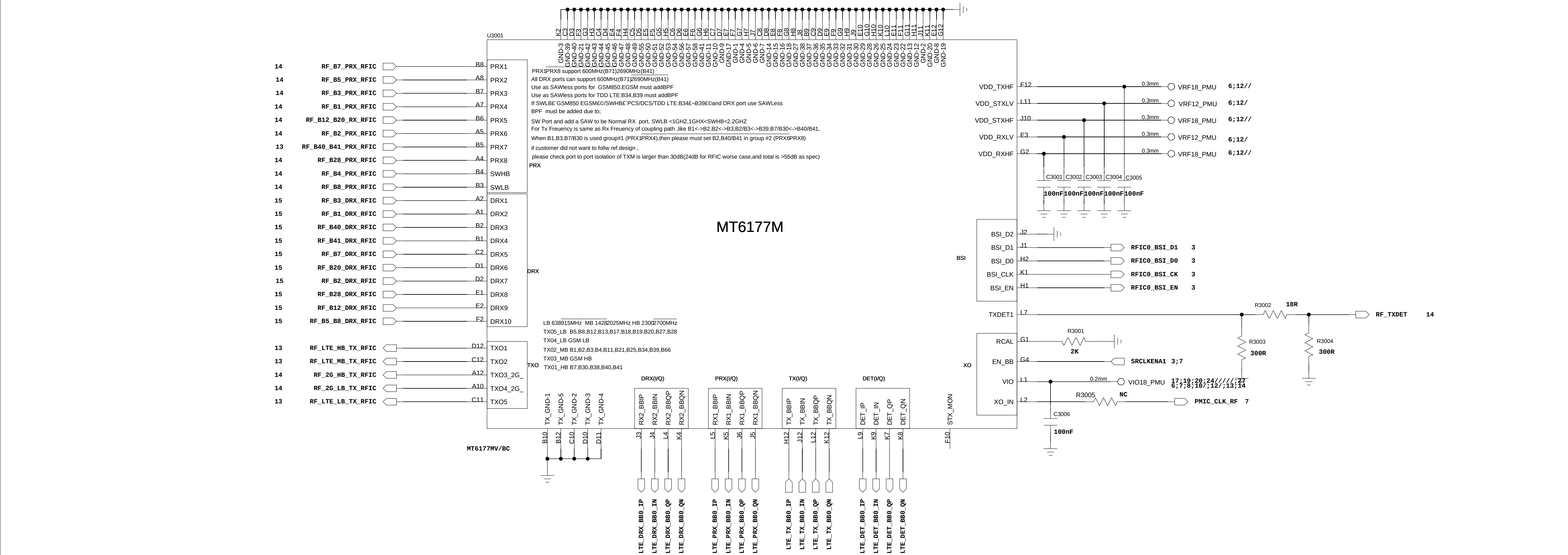
WIRELESS CHG

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

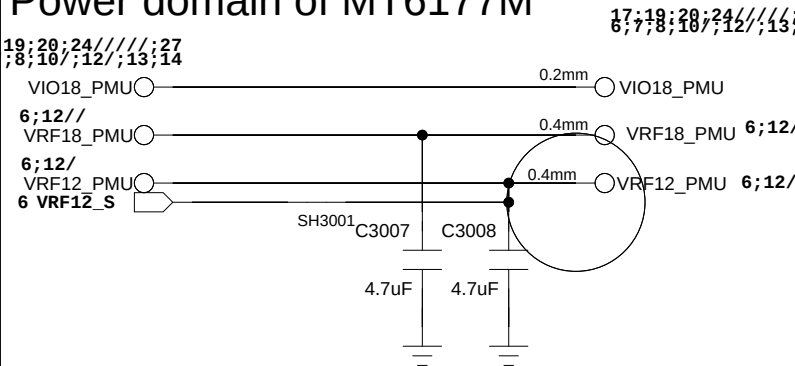
COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 28_POWER_THIRD-PARTY_IV		VERSION:	SHEET: 11 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6177M_PIN_OUT

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



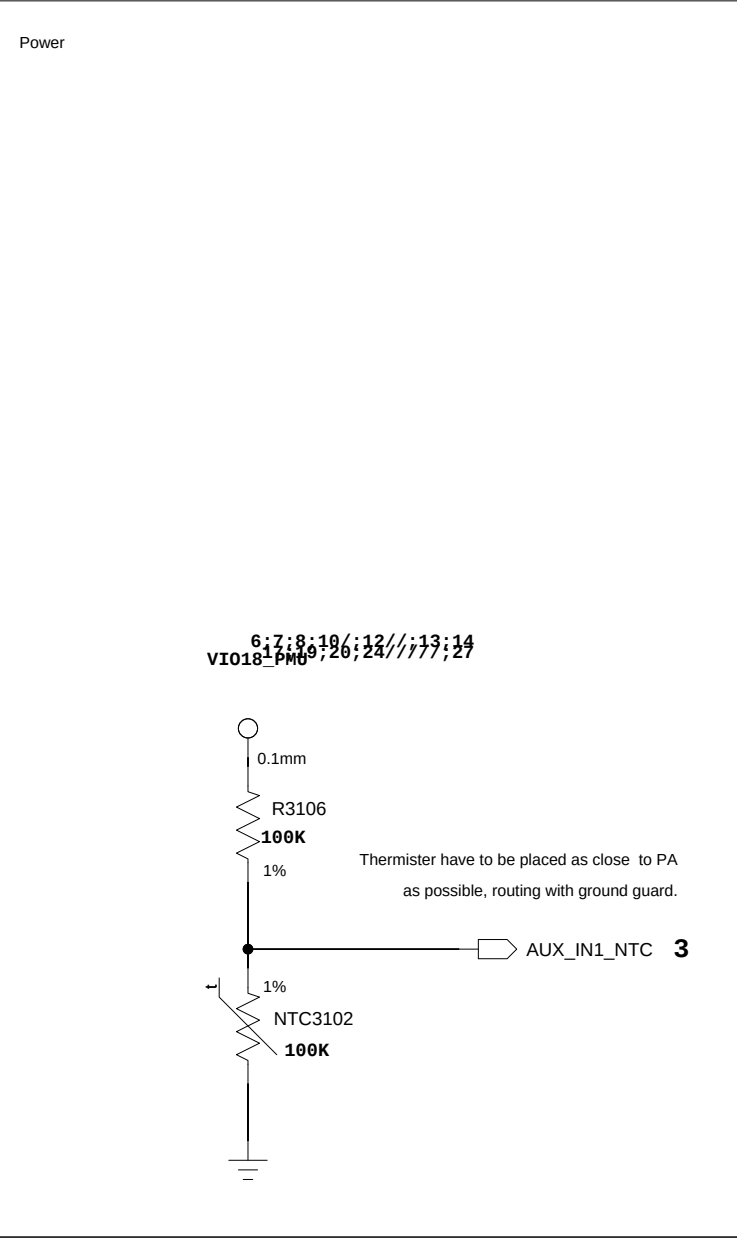
Power domain of MT6177M



Star Power

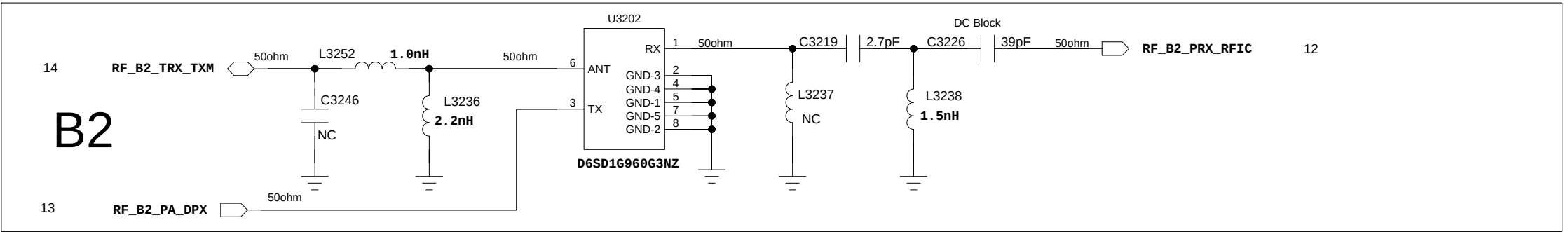
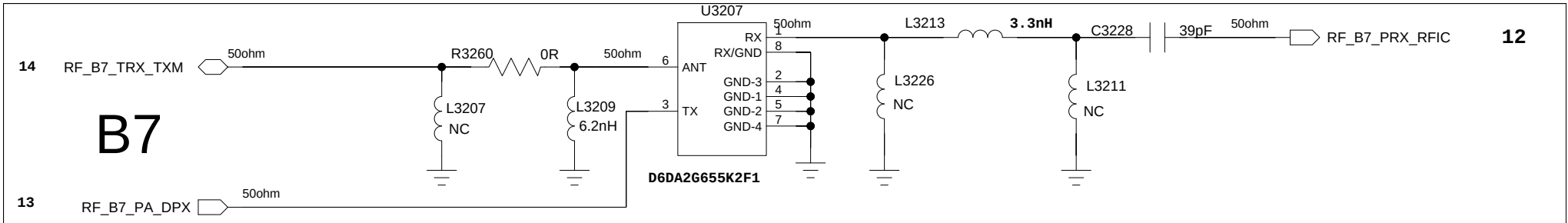
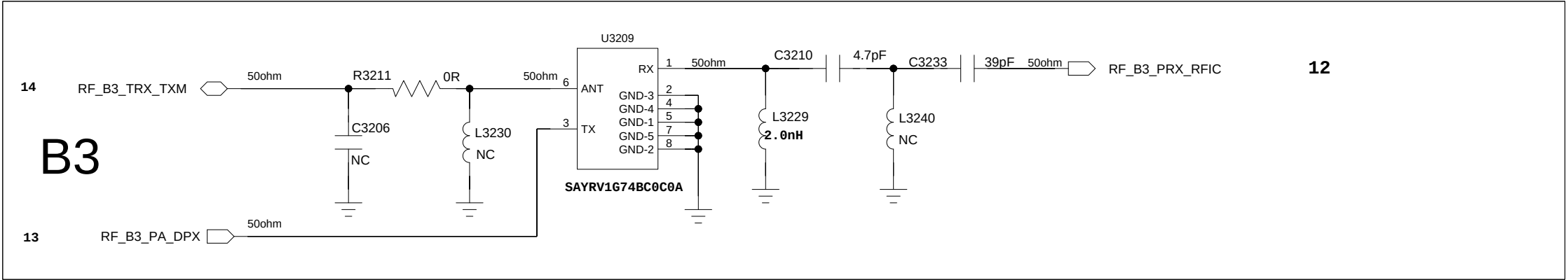
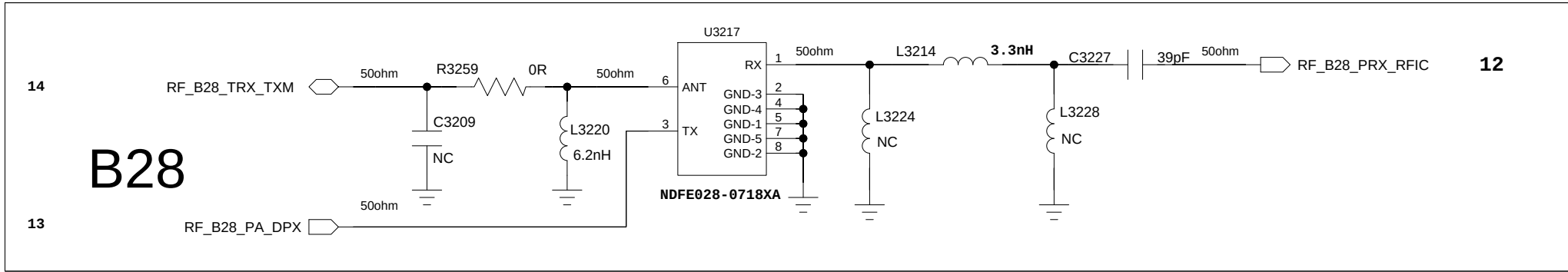
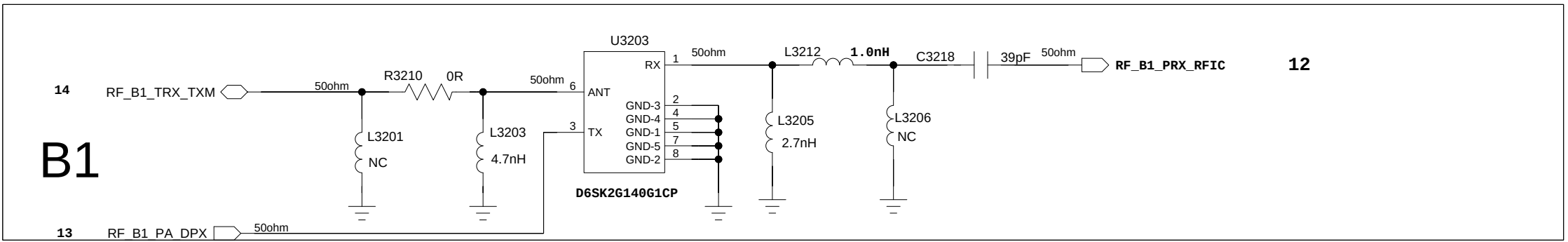
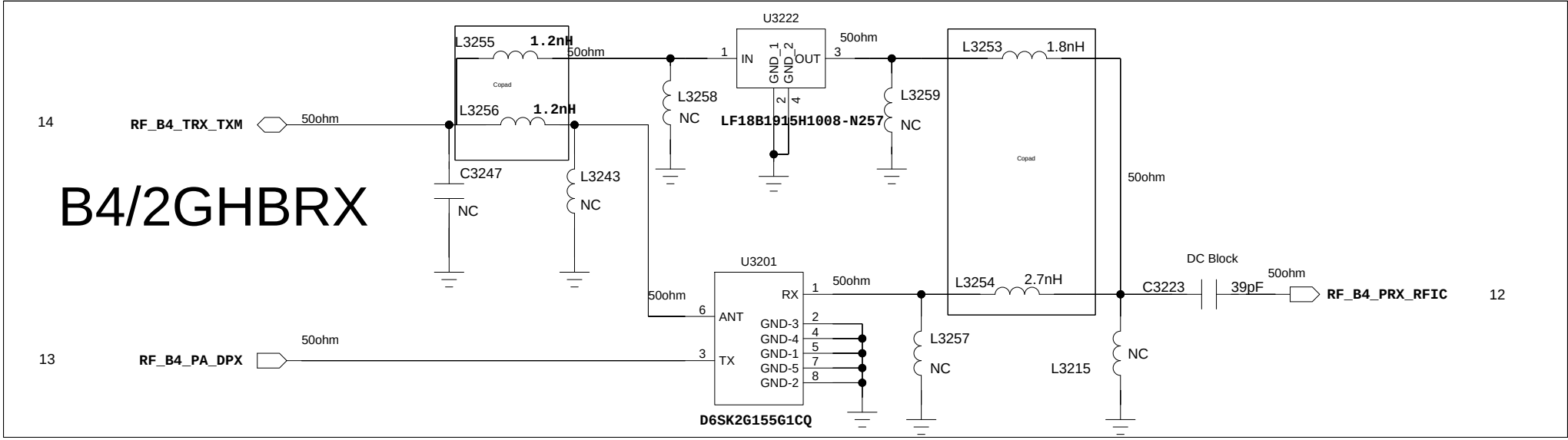
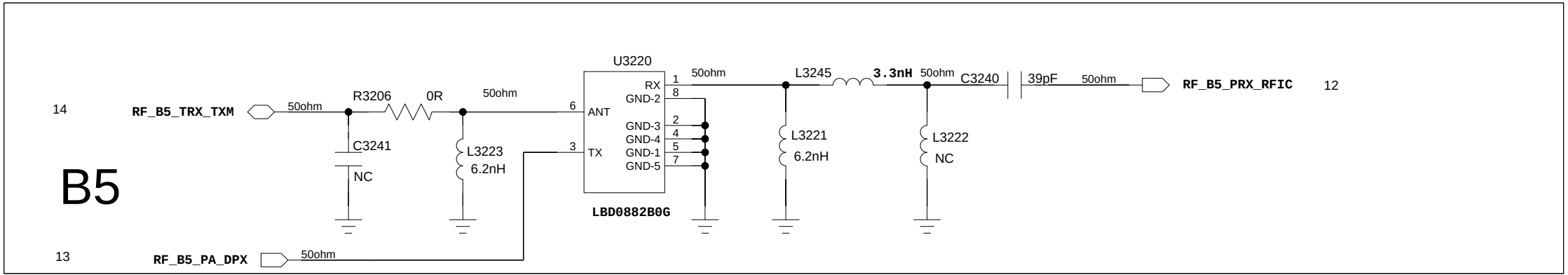
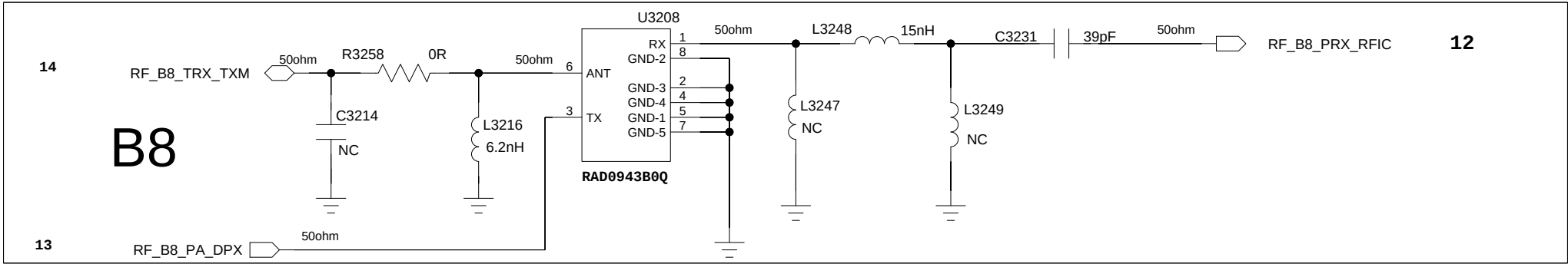
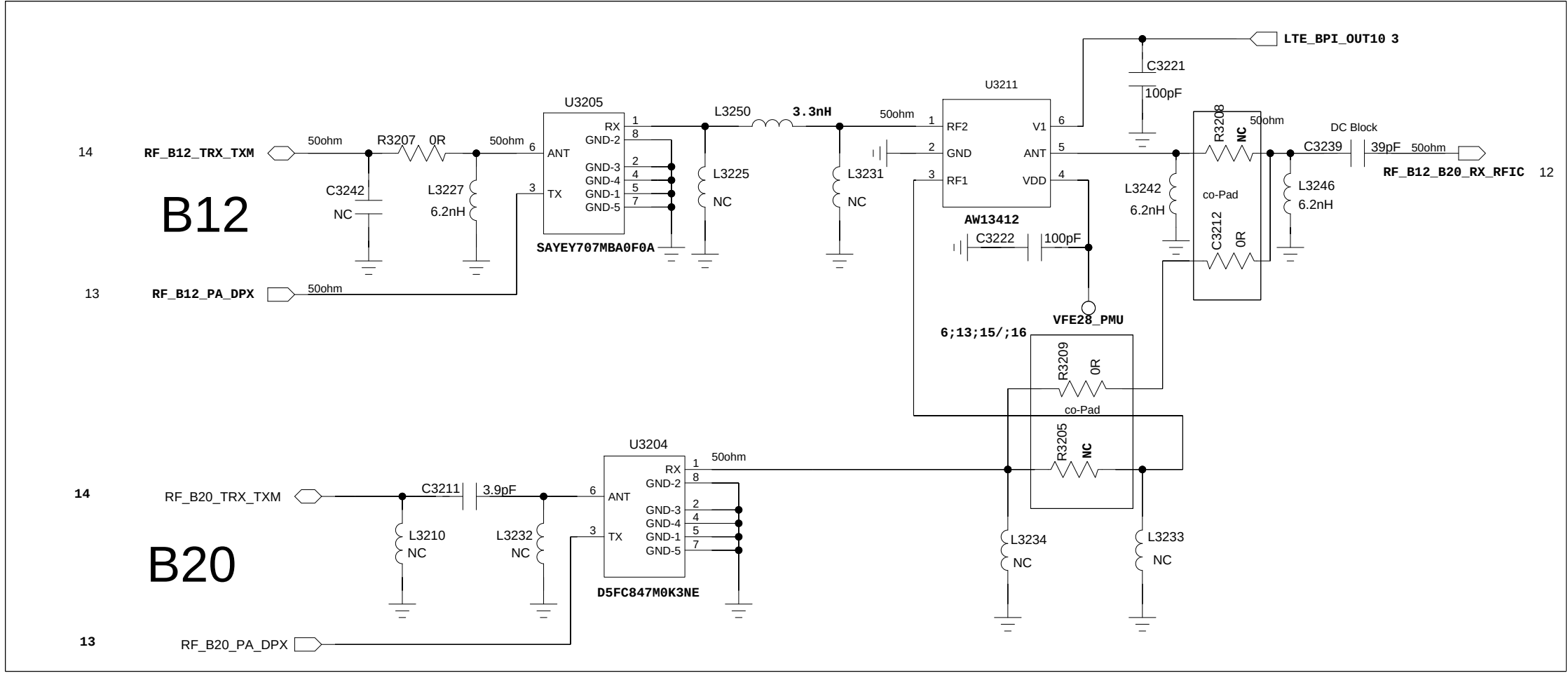
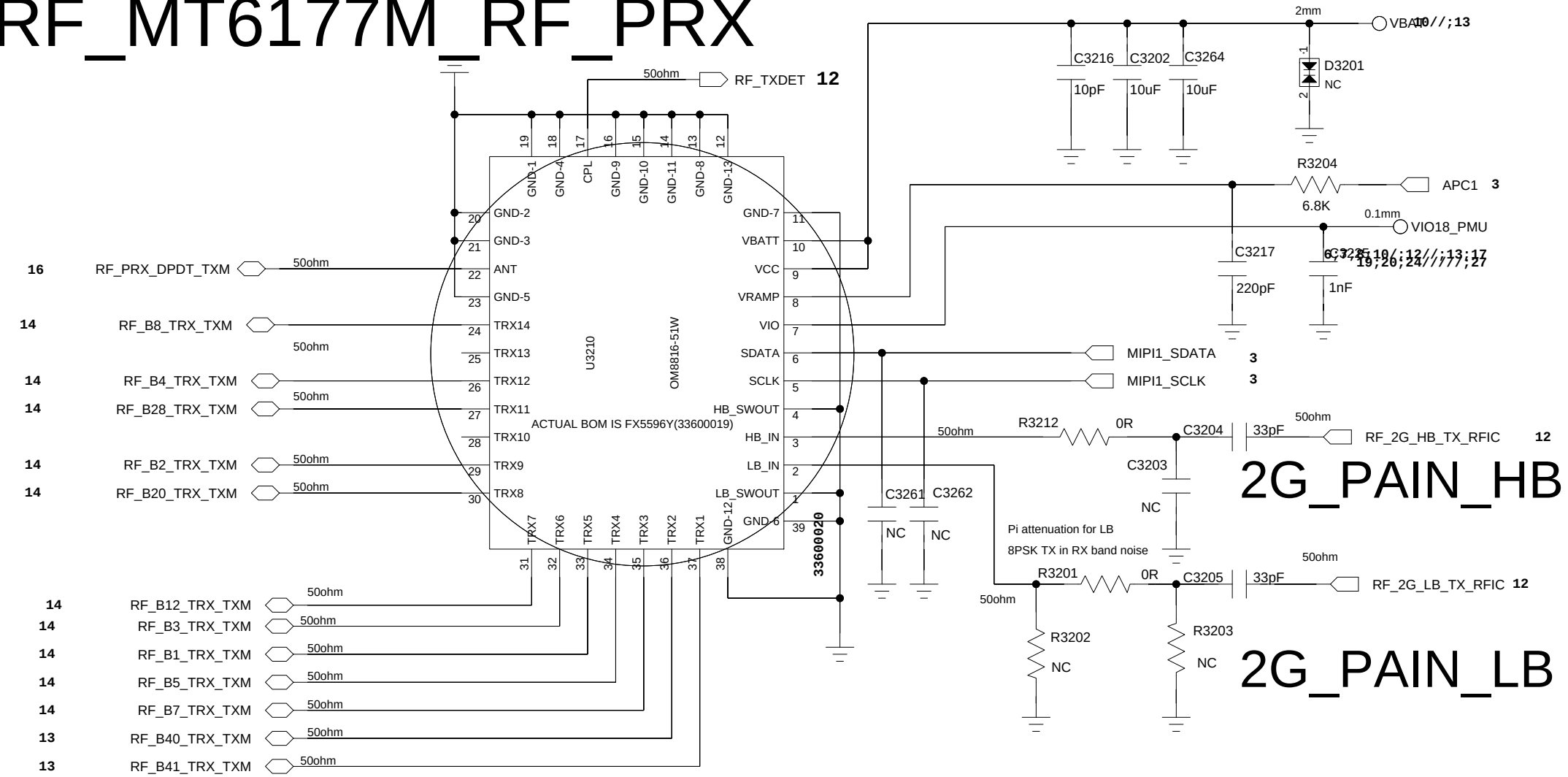
COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 30_RF_MT6177M_PIN_OUT		VERSION:	SHEET: 12 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



RF_MT6177M_RF_PRX

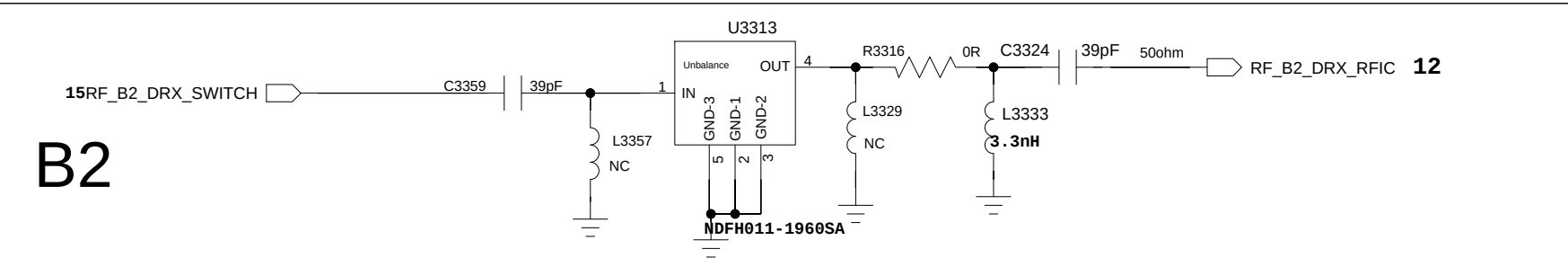
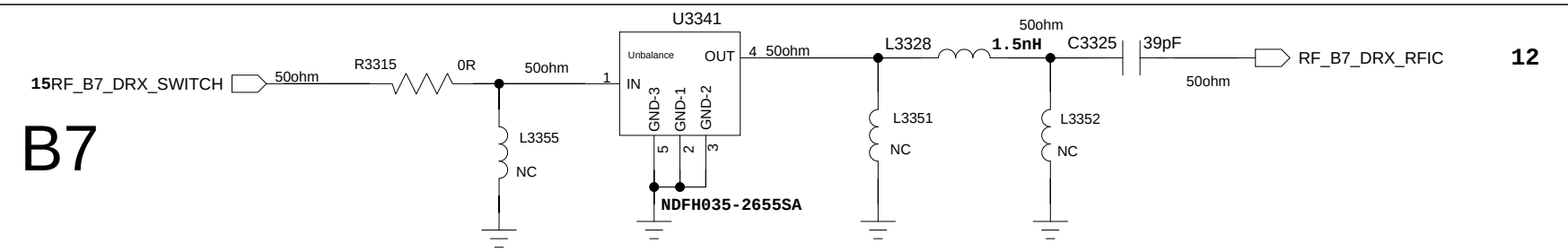
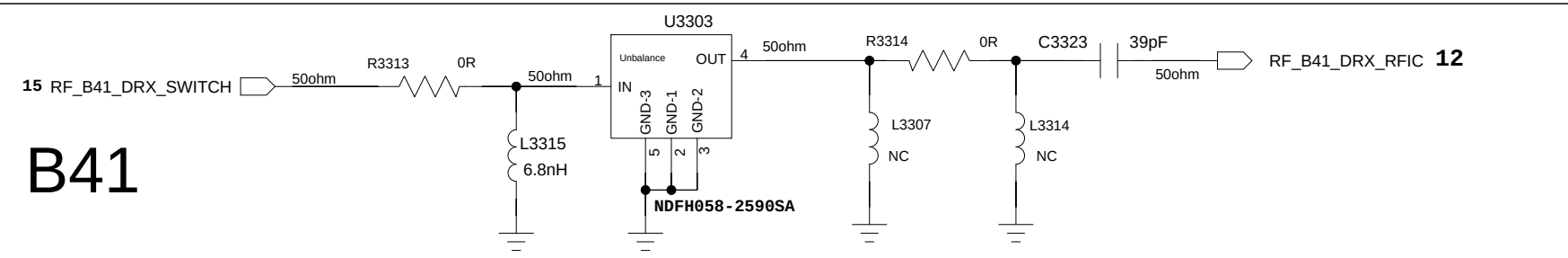
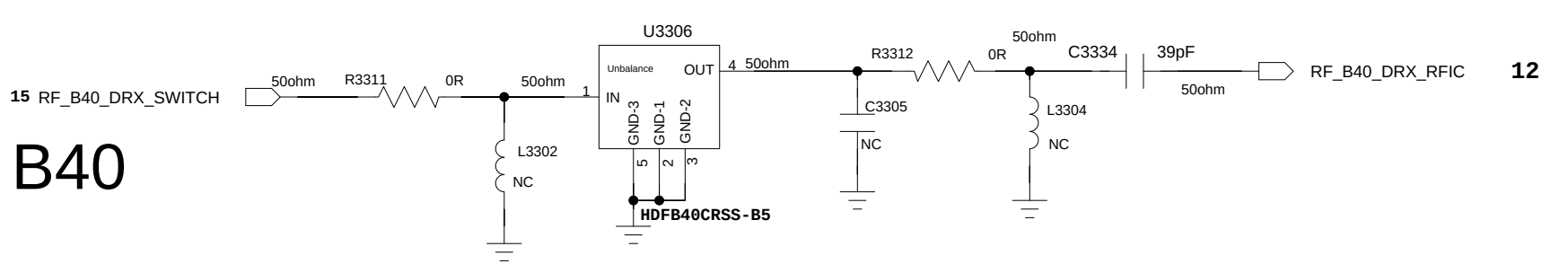
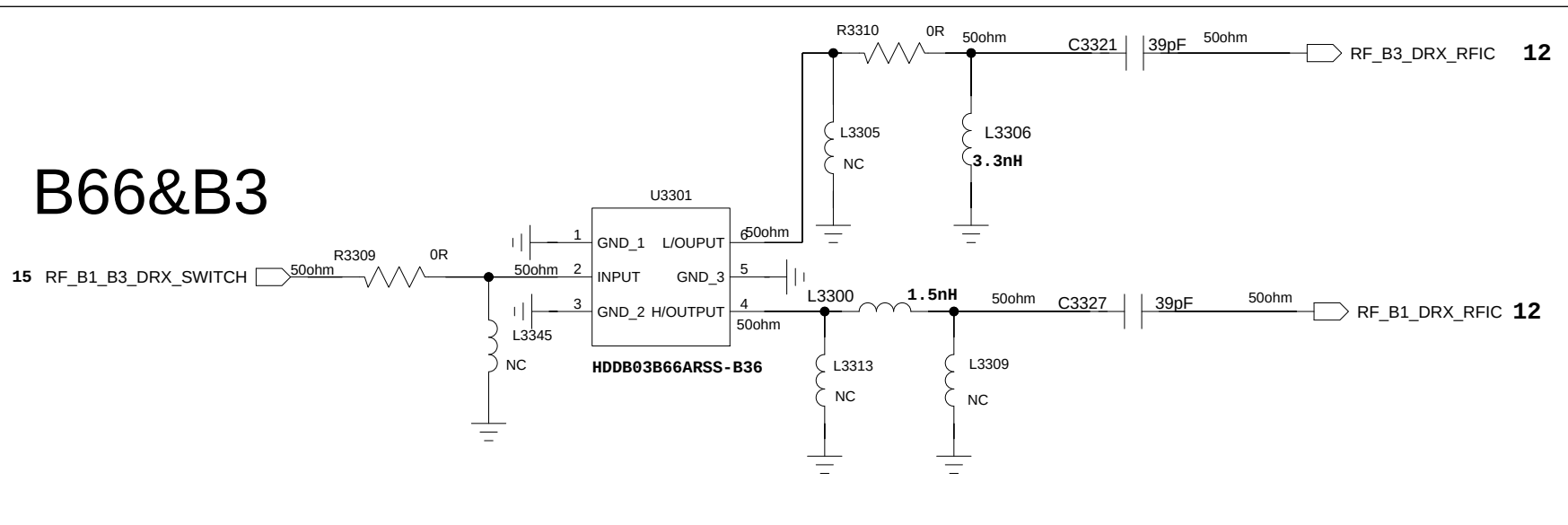
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL G85_STD_MAIN		Modified Date: 14/06/2023:18:02	
DRAWN		DATED		TITLE: 32_RF_MT6177M_RF_PRX		VERSION:	SHEET: 14 027
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

SP8T

The diagram illustrates the internal structure of the SP8T component, which is a U3312 AW13418. It features 16 pins: 1-15 are RF ports, and 16 is the control pin. The component is a 1:8 switch. The internal structure shows a switch controlled by the RF_DRX_DPDT_SP8T pin (pin 16). The switch is connected to the RF ports (pins 1-15) through 50ohm terminations. The control pin (16) is connected to the RF_DRX_DPDT_SP8T pin. The component is labeled SP8T.



VC1613 control logic					SP8T control logic										
VC1	VC2	RF1	RF2	RF3	SP6T control logic										
H	L	Y	N	N	V1	V2	V3	RF1	RF2	RF3	RF4	RF5	RF6	RF7	RF8
H	H	N	Y	N	L	L	L	Y	N	N	N	N	N	N	N
N	H	N	N	Y	L	L	H	N	Y	N	N	N	N	N	N
					L	H	L	N	N	Y	N	N	N	N	N
					L	H	H	N	N	N	Y	N	N	N	N
					H	L	L	N	N	N	N	Y	N	N	N
					H	L	H	N	N	N	N	N	Y	N	N
					H	H	L	N	N	N	N	N	N	Y	N
					H	H	H	N	N	N	N	N	N	N	Y

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LTR	ECO NO:	APPROVED:	DATE:

15

VFE28_PMU_DIV

○

R3350

OR

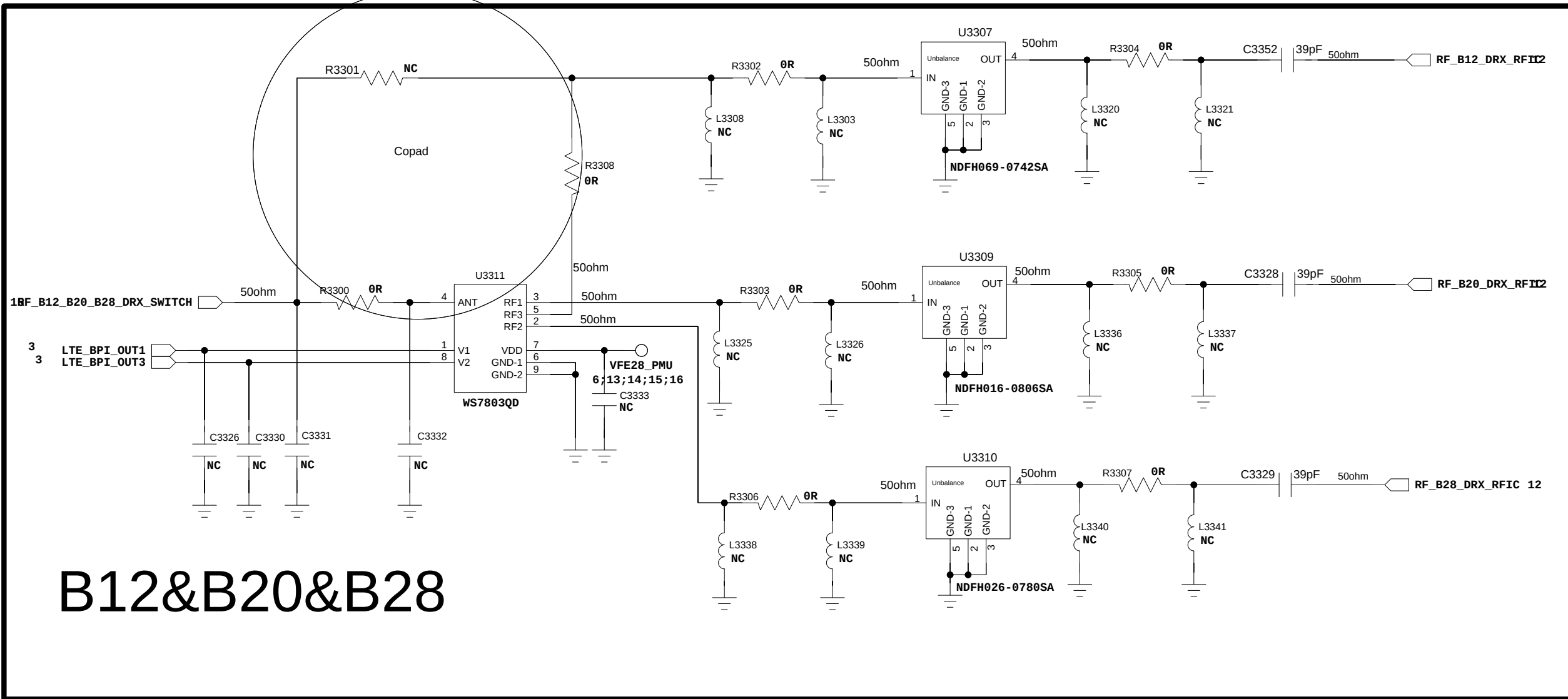
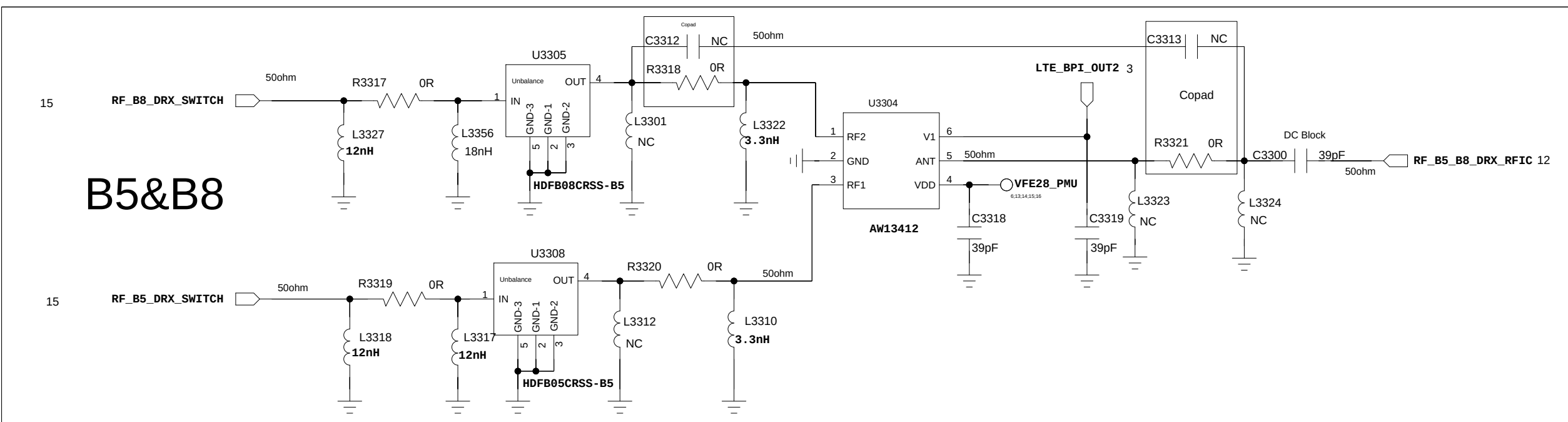
○

VFE28_PMU

0.2mm

Power

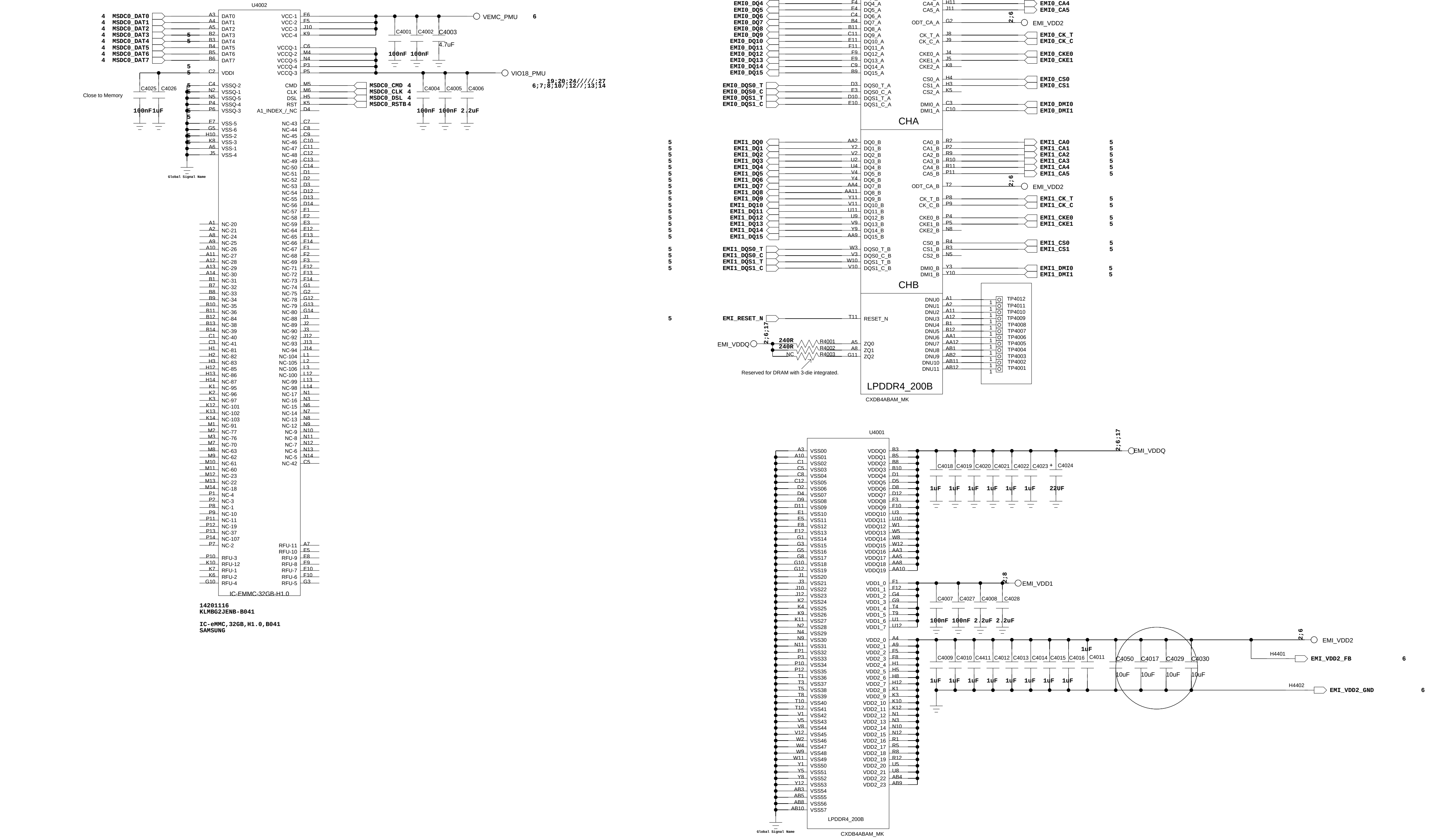
6;13;14;15;16



COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 33_RF_MT6177M_RF_DRX		VERSION:	SHEET: 15 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

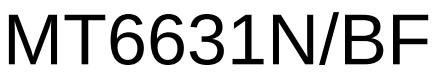
MEMORY_eMCP_LPDDR4X

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 40_MEMORY_EMCP_LPDDR4X		VERSION:	SHEET: 17 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Power domain

18 FM_AVDD28 0.2mm B5001 0R 0.25mm VCN28_PMU 6;18

C5019 1uF

18 AVDD18_GPS 0.2mm H5003 Star connection 0.25mm VCN18_PMU 6

NC-MMD-L8-3MIL C5012

18 AVDD18_WBT 0.2mm H5004 1uF

NC-MMD-L1-3MIL

Note: For MT6631 need to configure the VCN33_2_PMU to 2.8V by PMIC SW.

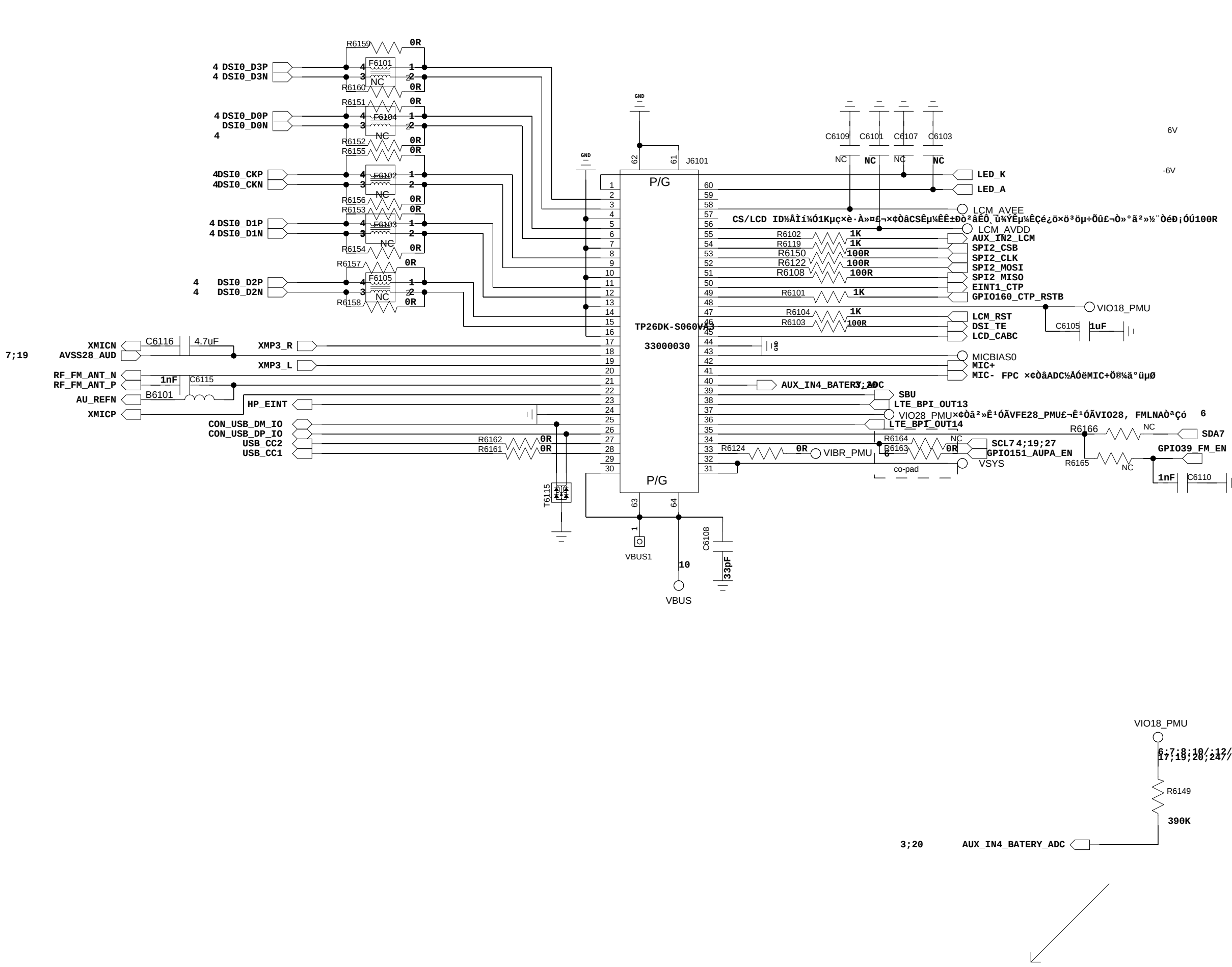
COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 14/06/2023:15:42	
DRAWN		DATED		TITLE: 50_CONNECTIVITY_MT6631		VERSION:	SHEET: 18 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_LCM_CTP_FP

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

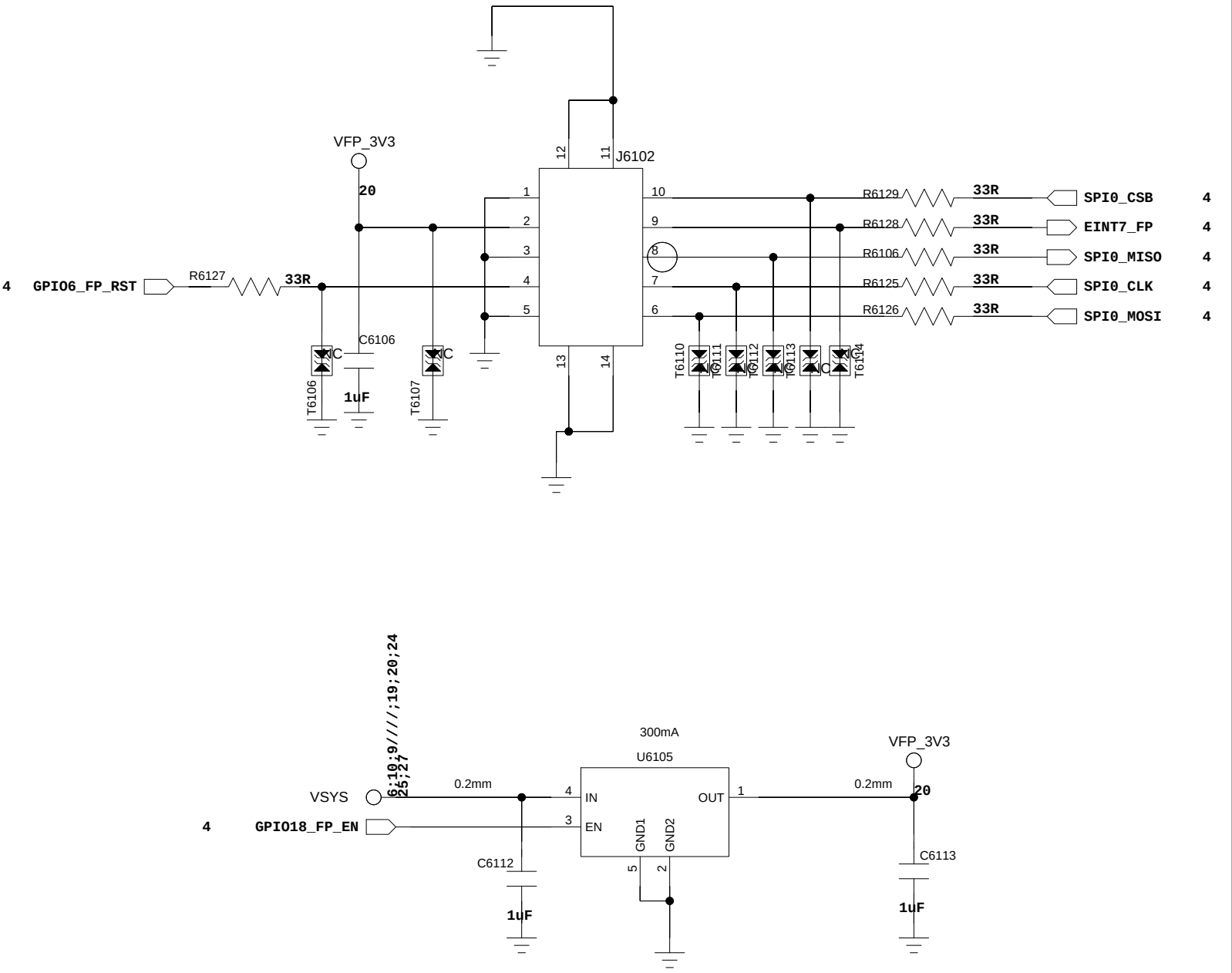
LCD_CTP_INTERFACE

CTP (NT36525) I2C address: 0X01 (Write:0x02, Read:0x03)



2. The distance is the shortest distance from package edge to edge.

FINGER PRINT



COMPANY: TRANSSION HOLDINGS

MODEL:G85_STD_MAIN

Modified Date: 12/06/2023:09:34

DRAWN

DATED

TITLE: 61_PERI_LCM_CTP_FP

CHECKED

DATED

< >

Confidentiality

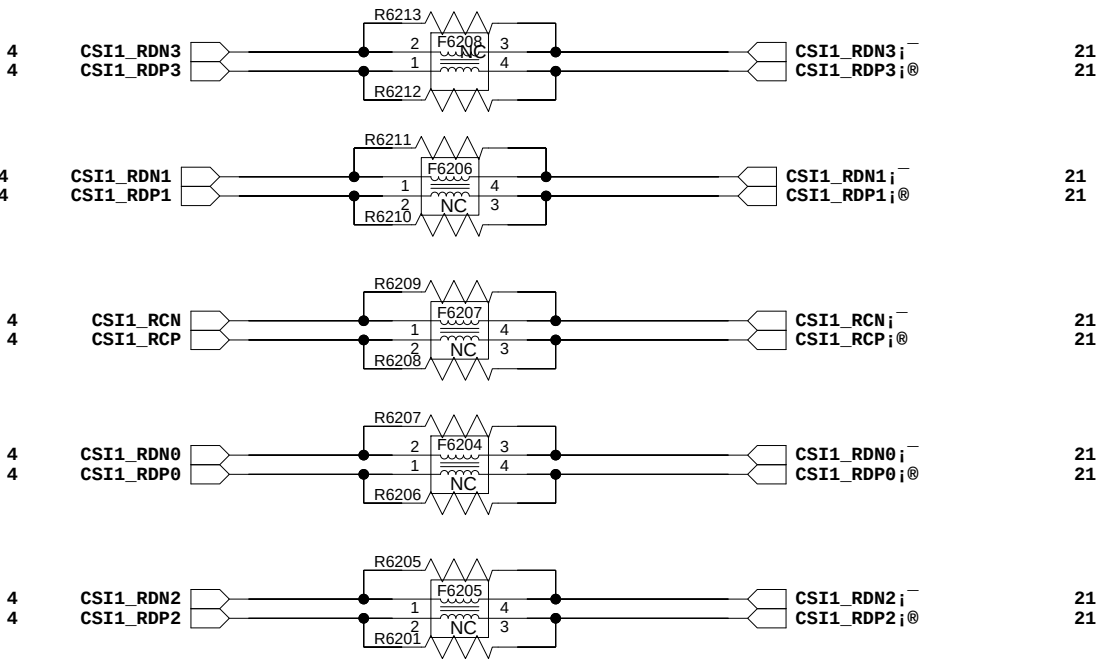
CONFIDENTIAL

VERSION:

SHEET: 20 OF 27

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

64M/50M/48M	R6203=0R	VCAMD_EXT1¹@μç
13M/16M/8M	R6203=NC	

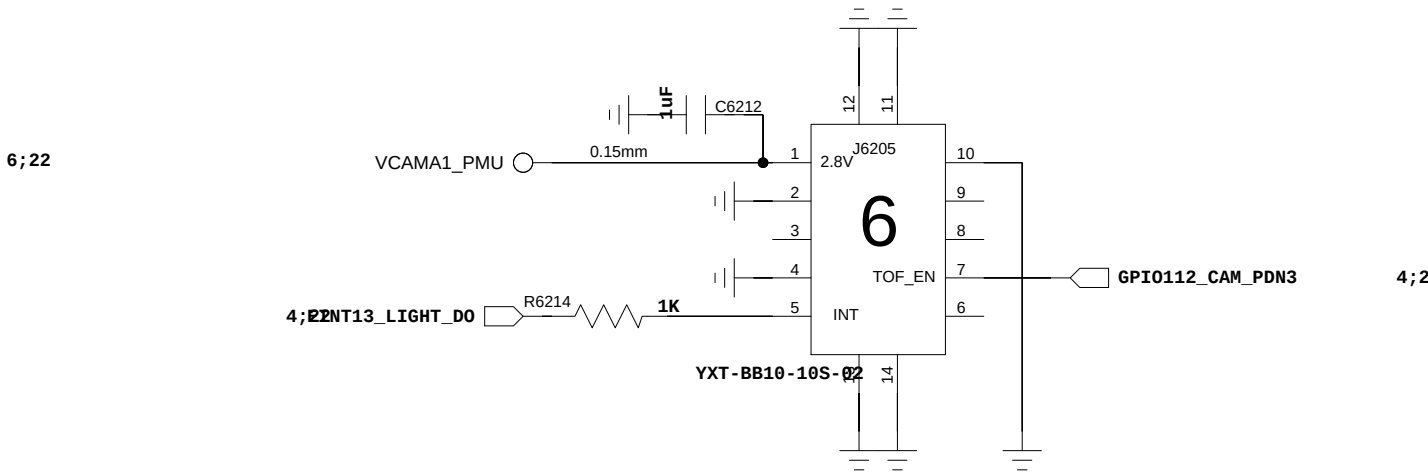
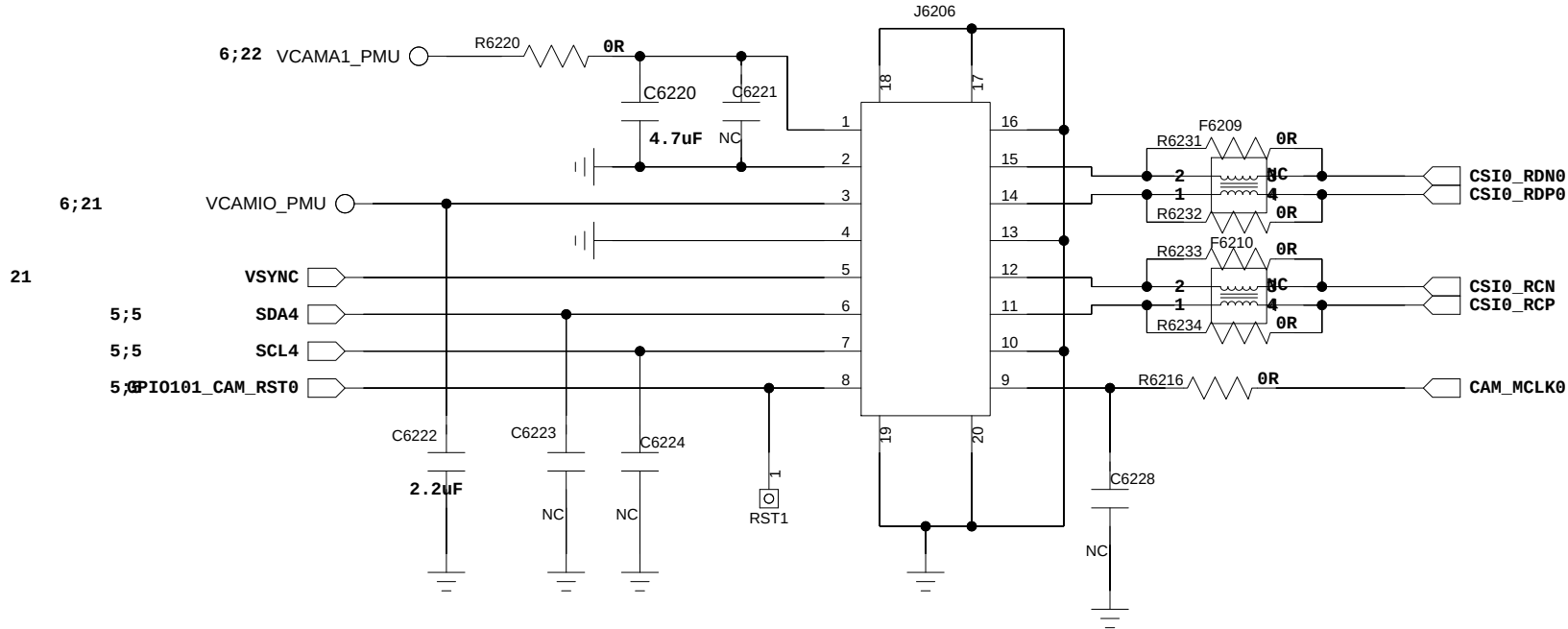


	SENSOR	I2C Address	VCAMA	VCAMD	VCAMIO	AFVDD	SAP
50M	H-5021P AA5021P1XX	Write 0x40 Read 0x41	2.8V @78mA	1.1V @598mA	1.8V@ 31mA	2.8V@ 103mA	17201826(0.7um)
	SS5KJ15Q03-FGX9	Write 0x20 Read 0x21	2.8V @67.7mA	1.05V @260.7mA	1.8V@ 4mA	2.8V@ 100mA	17201711(0.64um)
	H-1336-6A AA1336CNX-C20AHX	Write 0x40 Read 0x41	2.8V @44mA	1.1V @106mA	1.8V@ 6mA	2.8V@ 125mA	17201835
13M	SS5KJ36X03-FGX	Write 0x20 Read 0x21	2.8V @45mA	1.05V @160mA	1.8V@ 1.1mA	2.8V@ 125mA	17201836
	H846/A008A008B1C-B2CWX	Write 0x40 Read 0x41	2.8V @42mA	1.2V @67mA	1.8V@ 2mA	2.8V@ 123mA	17201833
	GA008A3-WA1XA	Write 0x62 Read 0x63	2.8V @60mA	1.2V @150mA	1.8V@ 10mA	2.8V@ 125mA	17201834
8M							

COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_I		VERSION:	SHEET: 21 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

2M



1 ãÃô

			VCAMA	PIN5			SAP
			2.8	1.7+0.2			17201729
1 ãÃô			2.8				17201744
			2.8				17201845

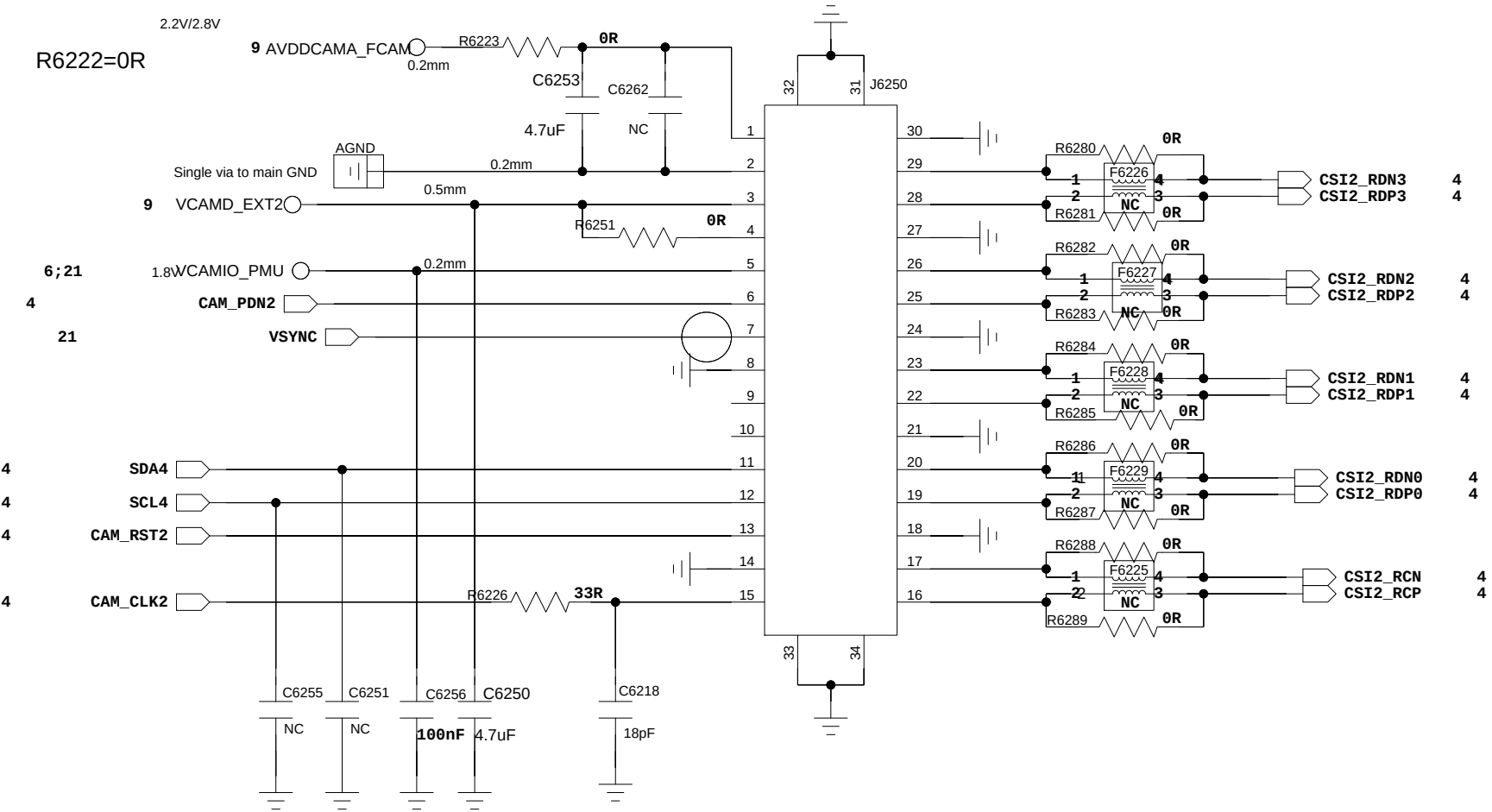
COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_II		VERSION:	SHEET: 22 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_CAMERA_III

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Front Camera (32M/16M/8M/5M)

32M	R6251=0R R6222=NC VCAMD_EXT1 ¹ @μç
5M/16M/8M	R6251=NC



	SENSOR	I2C Address	VCAMA	VCAMD	VCAMIO	SAP
32M	GC32E1-WA1XA	Write:0x94 Read:0x95	2.8V 43mA	1.2V 93.05mA	1.8V 4mA	
16M	HI-1631Q	Write:0x40 Read:0x41	2.8V 49mA	1.1V 115mA	1.8V 3mA	17201821
8M	GC8034-WC1X0	Write:0x6E Read:0x6F	2.8V 35mA	1.25V 140mA	1.8V 10mA	17201857
	GC08A3-MDD0	Write:0x24 Read:0x25	2.8V 60mA	1.2V 150mA	1.8V 10mA	17201829
	OV08856-GA4A-2	Write:0x6C Read:0x6D	2.8V 28mA	1.2V 95mA	1.8V 2.7mA	17201831
5M	GC5035-MCHDQ	0X6E(W)/0X6F(R)	2.8V 35mA	1.2V 80mA	1.8V 3mA	
	HI-556D(HYNIX)	0X50(W)/0X51(R)	2.8V 43mA	1.2V 46mA	1.8V 1mA	

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_III		VERSION:	SHEET: 23 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

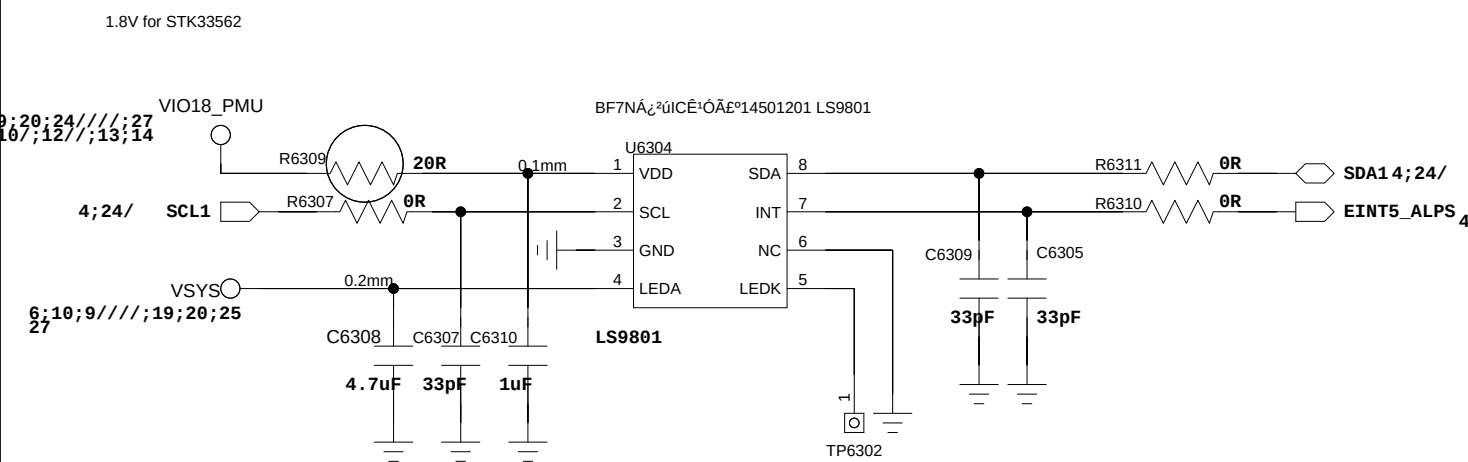
PERI_SENSORS

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

AL& PS Sensor

STK33562:I2C address: Write:0x8C, Read:0x8D

R6309μç×è²»½¨òé»»Îª0R£-Ö÷òªÎªÁÊ½μμÍμçÔ´øÀ´μÄμ×êÊÎÊîâ

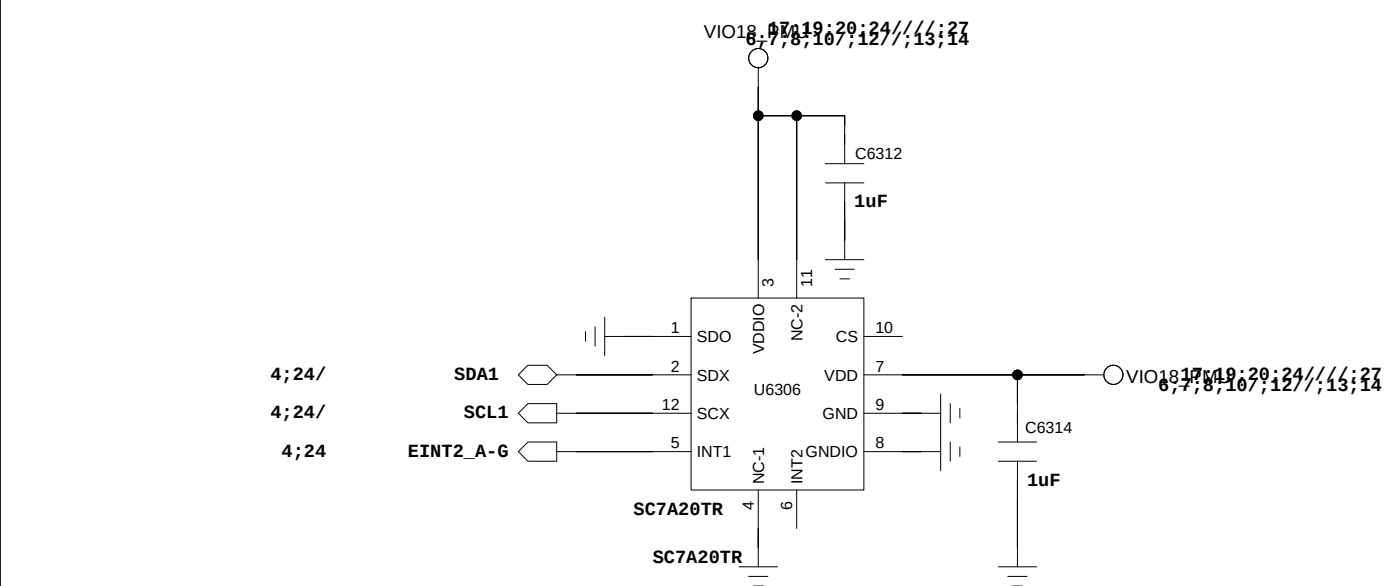


G-Sensor

MXC6655XA:ADDRESS:0x2A(Write)/0x2B(Read)

SC7A20:ADDRESS:0x30(Write)/0x31(Read)

MXC4005XC:ADDRESS:0x2A(Write)/0x2B(Read)



component	MXC6655XA	SC7A20	STK8BA50-S
C6302	NC	100nF	1uF
C6303	1uF	100nF	1uF

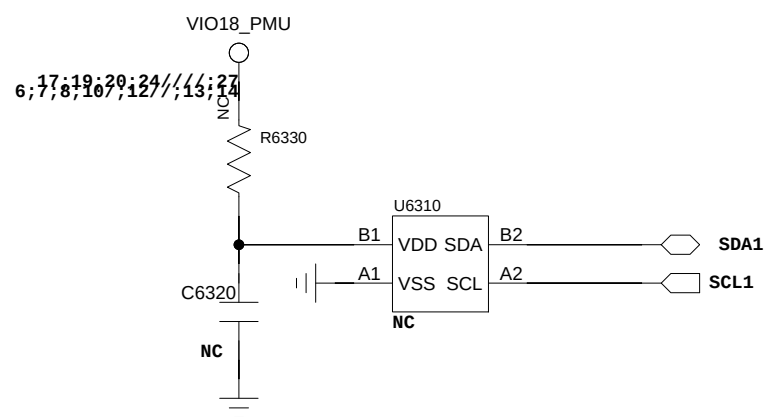
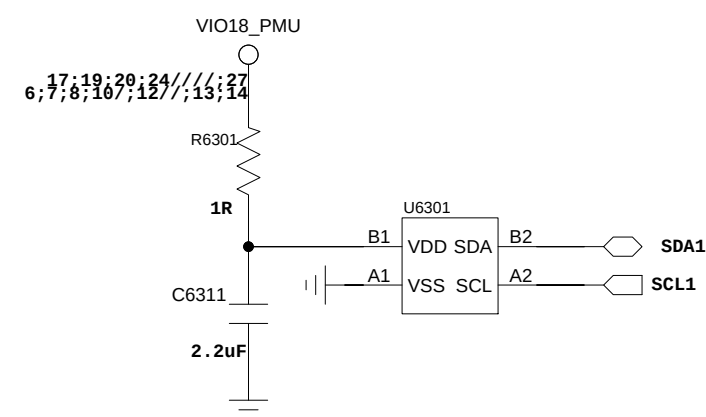
G-Sensor + Gyro Sensor

LSM6DS3TR-C:I2C address: Write:0xD4, Read:0xD5

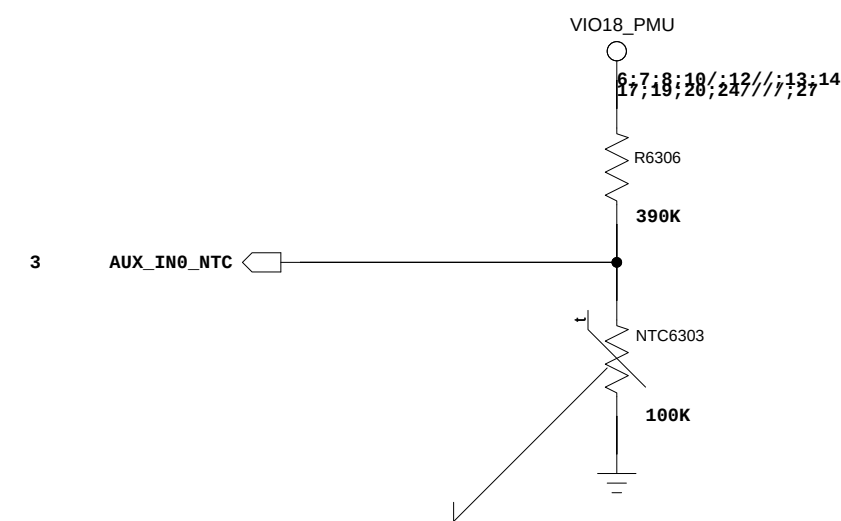
M-Sensor(COMPASS)

MMC5603:I2C ADDRESS:0x60(Write)/0x61(Read)

QMC6308:I2C ADDRESS:0x58(Write)/0x59(Read)



Thermistor to sense AP temperature



1. NTC6303 must keep a distance about 68 mm away from BB and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

COMPANY: TRANSSION HOLDINGS

MODEL:G85_STD_MAIN

Modified Date: **10/06/2023:19:27**

DRAWN

DATED

TITLE: 63_PERI_SENSORS

CHECKED

DATED

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Confidentiality

CONFIDENTIAL

VERSION:

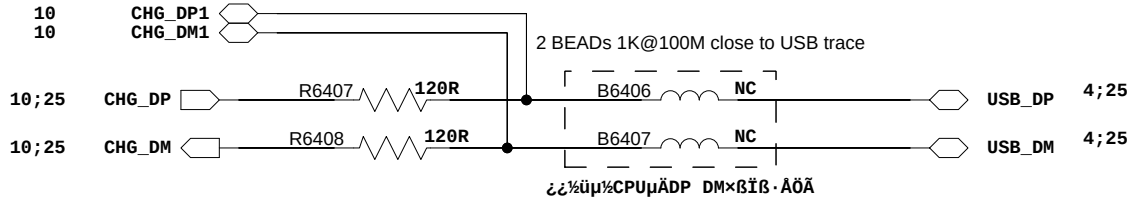
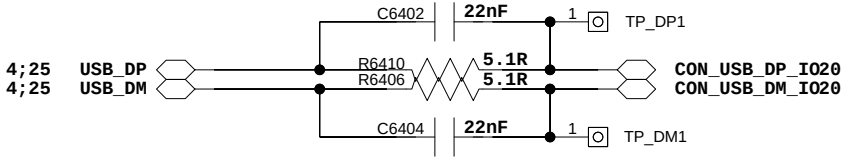
SHEET: 24 OF 27

PERI_EXCH_IO

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

RFC

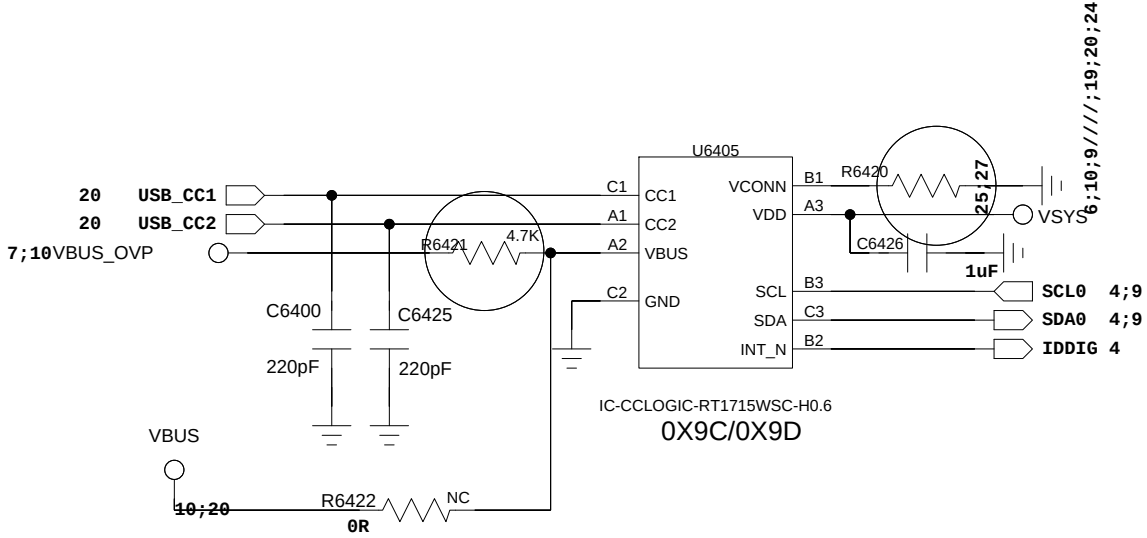
Put it on the trace,do not by-passed Place on battery side outside shielding case



18W IC

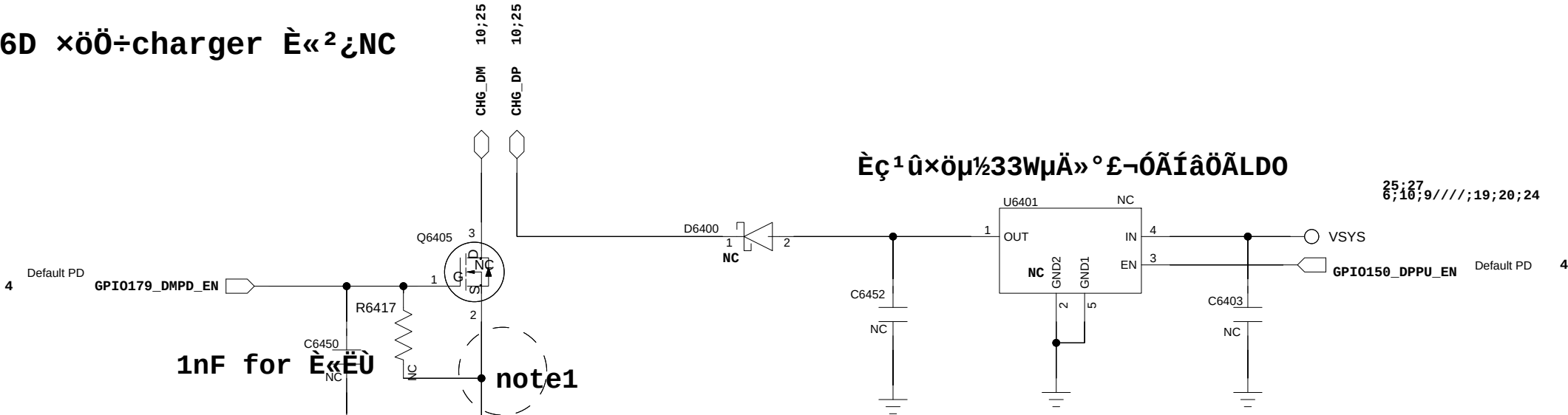
18W IC	T6405	R6407	R6408	18W IC
OTHERS	RES-->0R	120R	120R	18W IC

CC Logic



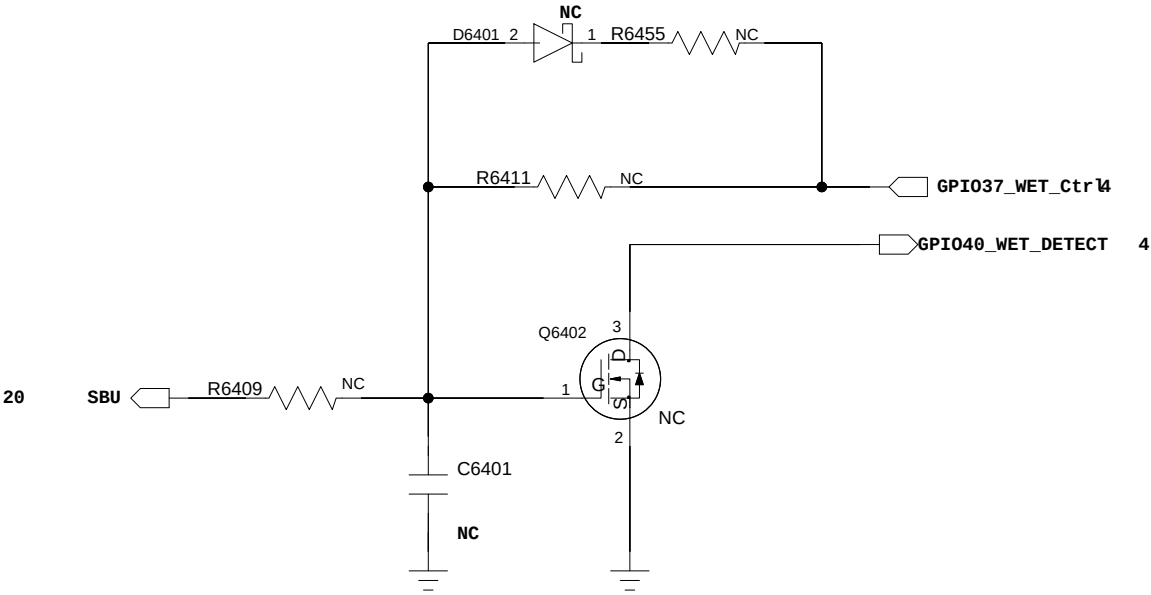
HVDCP TA

SGM41516D charger

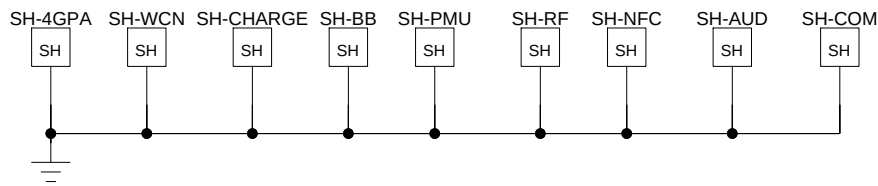


note1: RT9467/ETA69630A-QC00A-μçÂ · ; ÉÊ¹ÓÄRT9470D/SGM41516DμÄÊ±°ò¶p%«¹ÜNCÖ±½ÓÏÂGND ; É

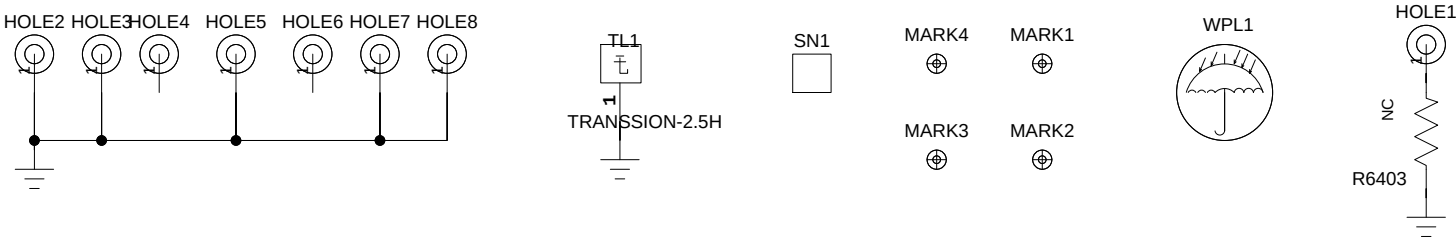
WET detect



SHIELDING



LOGO&SN&MARK

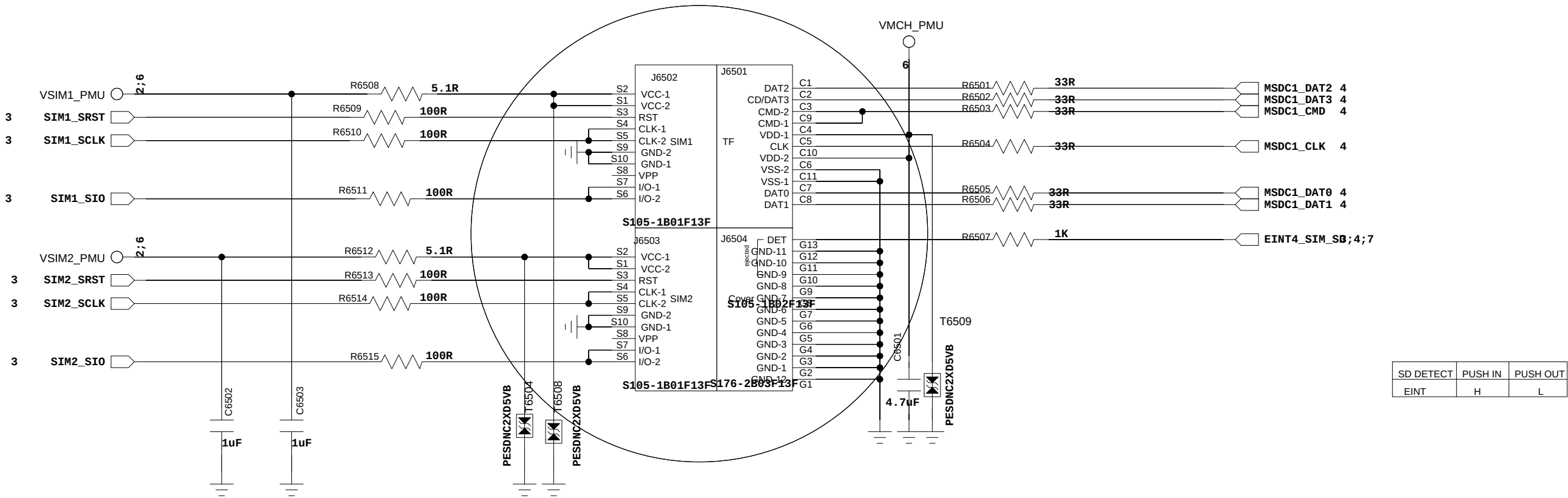


COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 13/06/2023:11:15	
DRAWN		DATED		TITLE: 64_PERI_EXCH_IO		VERSION:	SHEET: 25 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_SIM_SD_KEYPAD

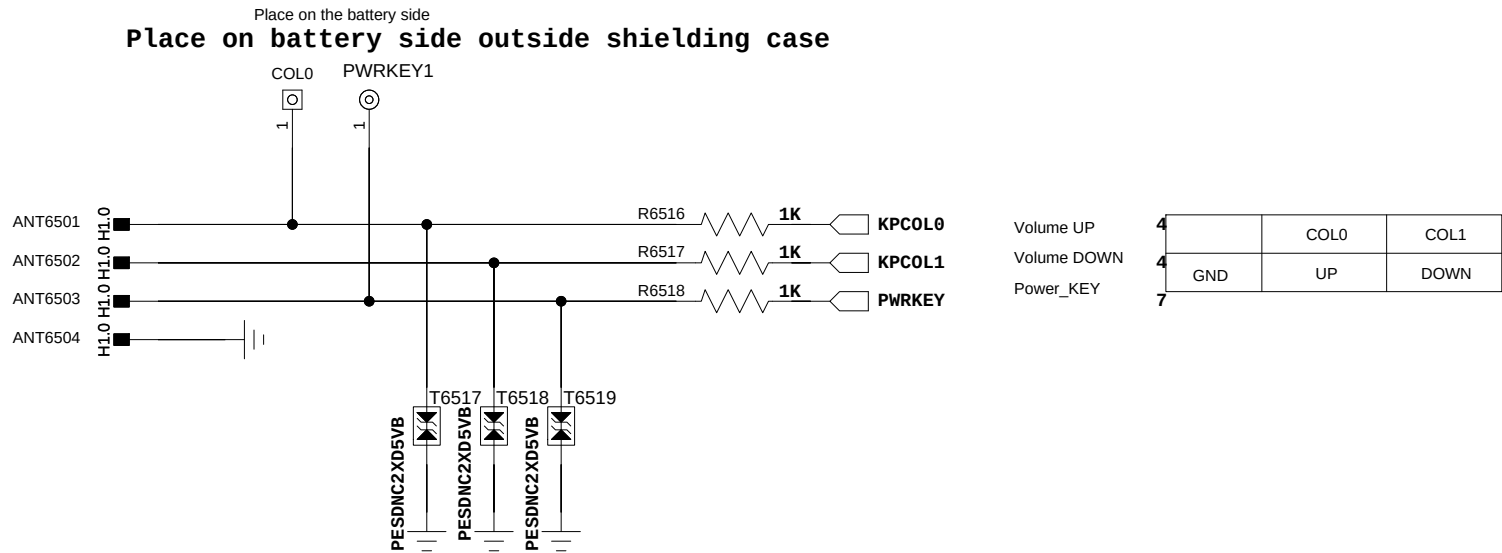
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

SIM1/2-SD-CARD



SD POWER CONTROL

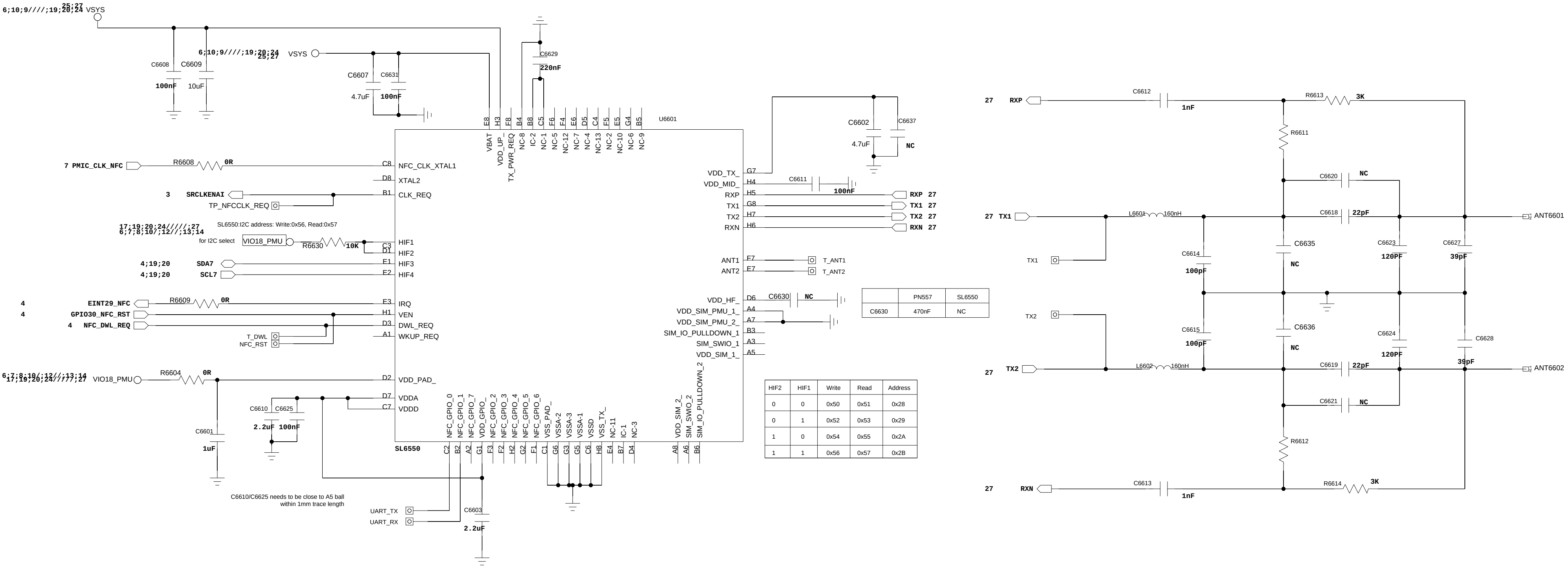
SIDEKEY



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 65_PERI_SIM_SD_KEYPAD		VERSION:	SHEET: 26 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_NFC

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 66_PERI_NFC		VERSION:	SHEET: 27 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		